Business_Case_Delhivery_Jiyansh_Garg

March 6, 2026

1 Business Case - DELHIVERY

1.1 Introduction

Delhivery is one of India's largest integrated logistics and supply chain companies. It operates across multiple transportation modes such as Full Truck Load (FTL) and Carting, managing shipments between various source and destination centers across the country.

In logistics operations, efficiency in delivery time, route optimization, and forecasting accuracy plays a critical role in reducing costs and improving customer satisfaction. The dataset provided contains detailed trip-level and segment-level delivery information, including actual time taken, estimated time using OSRM (Open Source Routing Machine), distances, and timestamps.

However, the dataset is raw and structured at a segment level, where a single trip may be divided into multiple rows. Before this data can be used for forecasting models or business analysis, it needs to be cleaned, structured, and transformed.

1.2 Problem Statement

The primary problem in this case study is to transform raw, segment-level logistics data into a clean, structured, and trip-level dataset that can be used for meaningful analysis and forecasting.

Currently, the data:

- Contains multiple rows for a single trip
- Includes cumulative and segment-level time and distance values
- Has timestamp fields that require conversion
- May contain missing values and outliers
- Lacks engineered features necessary for predictive modeling

The challenge is to process this data in a way that ensures accuracy, consistency, and usability for downstream analytics and machine learning tasks.

1.3 Objective of the Case Study

The main objectives of this business case study are:

1. To clean and preprocess the dataset by handling missing values, correcting data types, and converting time columns into proper datetime formats.
2. To merge and aggregate segment-level rows into meaningful segment-level and trip-level summaries using appropriate grouping and aggregation techniques.
3. To perform feature engineering by extracting useful information such as trip duration, month, year, city, state, and time-based features.
4. To compare actual delivery time with OSRM estimated time to evaluate route efficiency and estimation accuracy.
5. To detect and handle outliers in numerical features using statistical techniques like the IQR method.
6. To prepare the dataset for modeling by applying one-hot encoding to categorical variables and scaling numerical variables.
7. To derive actionable business insights that can help Delhivery improve operational efficiency, reduce delays, and optimize logistics planning.

1.4 Business Relevance

This case study helps understand how logistics data can be transformed into meaningful insights. By analyzing actual versus estimated time, identifying busy corridors, and examining delivery patterns across states and routes, Delhivery can:

- Improve time forecasting accuracy
- Optimize route planning
- Reduce operational inefficiencies
- Enhance customer satisfaction

The structured dataset produced after feature engineering can also be used by the data science team to build reliable forecasting models for delivery time prediction.

1.5 Importing Libraries and Dataset

[ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

[225]:
```
# dataset link:-
# https://docs.google.com/spreadsheets/d/e/
↪2PACX-1vRL2Humv8jYZRFckqOjGauGw-pvFOGGqQyJWVycx729zJj-tDPh2fKsbAPjZG_TJE7eHTCzDGjxTS5R/
↪pub?gid=1814535743&single=true&output=csv

# colab notebook link:-
```

```
# https://drive.google.com/file/d/111lH3FMCcqrlX_dVolKWCTrrFx9xuxHS/view?
↪usp=sharing
```

[ ]:
```
url = "https://docs.google.com/spreadsheets/d/e/
↪2PACX-1vRL2Humv8jYZRFckqOjGauGw-pvFOGGqQyJWVycx729zJj-tDPh2fKsbAPjZG_TJE7eHTCzDGjxTS5R/
↪pub?gid=1814535743&single=true&output=csv"
delhivery_data = pd.read_csv(url)
print(delhivery_data)
```

```
            data   trip_creation_time  \
0       training   2018-09-20 2:35:36
1       training   2018-09-20 2:35:36
2       training   2018-09-20 2:35:36
3       training   2018-09-20 2:35:36
4       training   2018-09-20 2:35:36
…           …                   …
144862  training  2018-09-20 16:24:28
144863  training  2018-09-20 16:24:28
144864  training  2018-09-20 16:24:28
144865  training  2018-09-20 16:24:28
144866  training  2018-09-20 16:24:28

                                     route_schedule_uuid route_type  \
0       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
3       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
…                                                     …        …
144862  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144863  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144864  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144865  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144866  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting

                      trip_uuid source_center                 source_name  \
0       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
…                           …             …                          …
144862  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144863  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144864  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144865  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144866  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
```

```
       destination_center               destination_name        od_start_time  \
0            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
1            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
2            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
3            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
4            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
…                     …                            …                  …
144862       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144863       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144864       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144865       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144866       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28

        …     cutoff_timestamp  actual_distance_to_destination  actual_time  \
0       …   2018-09-20 4:27:55                       10.435660           14
1       …   2018-09-20 4:17:55                       18.936842           24
2       …   2018-09-20 4:01:20                       27.637279           40
3       …   2018-09-20 3:39:57                       36.118028           62
4       …   2018-09-20 3:33:55                       39.386040           68
…     …                  …                            …           …
144862  …  2018-09-20 21:57:20                       45.258278           94
144863  …  2018-09-20 21:31:18                       54.092531          120
144864  …  2018-09-20 21:11:18                       66.163591          140
144865  …  2018-09-20 20:53:19                       73.680667          158
144866  …  2018-09-20 16:24:28                       70.039010          426

        osrm_time osrm_distance    factor  segment_actual_time  \
0              11       11.9653  1.272727                   14
1              20       21.7243  1.200000                   10
2              28       32.5395  1.428571                   16
3              40       45.5620  1.550000                   21
4              44       54.2181  1.545455                    6
…           …           …       …                   …
144862         60       67.9280  1.566667                   12
144863         76       85.6829  1.578947                   26
144864         88       97.0933  1.590909                   20
144865         98      111.2709  1.612245                   17
144866         95       88.7319  4.484211                  268

        segment_osrm_time  segment_osrm_distance  segment_factor
0                      11                11.9653        1.272727
1                       9                 9.7590        1.111111
2                       7                10.8152        2.285714
3                      12                13.0224        1.750000
4                       5                 3.9153        1.200000
…                   …                   …             …
144862                 12                 8.1858        1.000000
```

```
144863                 21                 17.3725        1.238095
144864                 34                 20.7053        0.588235
144865                 27                 18.8885        0.629630
144866                  9                  8.8088       29.777778

[144867 rows x 24 columns]
```

[ ]:
```python
# Creating copy of dataset to preserve the originality
df = delhivery_data.copy()
df
```

[ ]:
```
            data   trip_creation_time  \
0       training   2018-09-20 2:35:36
1       training   2018-09-20 2:35:36
2       training   2018-09-20 2:35:36
3       training   2018-09-20 2:35:36
4       training   2018-09-20 2:35:36
…          …                  …
144862  training  2018-09-20 16:24:28
144863  training  2018-09-20 16:24:28
144864  training  2018-09-20 16:24:28
144865  training  2018-09-20 16:24:28
144866  training  2018-09-20 16:24:28

                                     route_schedule_uuid route_type  \
0       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
3       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4       thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
…                                                  …          …
144862  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144863  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144864  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144865  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting
144866  thanos::sroute:f0569d2f-4e20-4c31-8542-67b86d5…    Carting

                      trip_uuid source_center                 source_name  \
0       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4       trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
…                           …             …                          …
144862  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144863  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144864  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
```

```
144865  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)
144866  trip-153746066843555182  IND131028AAB  Sonipat_Kundli_H (Haryana)

       destination_center               destination_name        od_start_time  \
0            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
1            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
2            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
3            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
4            IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)   2018-09-20 3:21:32
…                   …                                …                  …
144862       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144863       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144864       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144865       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28
144866       IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)  2018-09-20 16:24:28

        …     cutoff_timestamp  actual_distance_to_destination  actual_time  \
0       …   2018-09-20 4:27:55                       10.435660           14
1       …   2018-09-20 4:17:55                       18.936842           24
2       …   2018-09-20 4:01:20                       27.637279           40
3       …   2018-09-20 3:39:57                       36.118028           62
4       …   2018-09-20 3:33:55                       39.386040           68
…     …                 …                             …          …
144862  …  2018-09-20 21:57:20                       45.258278           94
144863  …  2018-09-20 21:31:18                       54.092531          120
144864  …  2018-09-20 21:11:18                       66.163591          140
144865  …  2018-09-20 20:53:19                       73.680667          158
144866  …  2018-09-20 16:24:28                       70.039010          426

        osrm_time osrm_distance    factor  segment_actual_time  \
0              11       11.9653  1.272727                   14
1              20       21.7243  1.200000                   10
2              28       32.5395  1.428571                   16
3              40       45.5620  1.550000                   21
4              44       54.2181  1.545455                    6
…           …           …       …                   …
144862         60       67.9280  1.566667                   12
144863         76       85.6829  1.578947                   26
144864         88       97.0933  1.590909                   20
144865         98      111.2709  1.612245                   17
144866         95       88.7319  4.484211                  268

        segment_osrm_time  segment_osrm_distance  segment_factor
0                      11                11.9653        1.272727
1                       9                 9.7590        1.111111
2                       7                10.8152        2.285714
3                      12                13.0224        1.750000
```

```
4                        5                  3.9153        1.200000
…                      …                    …             …
144862                  12                  8.1858        1.000000
144863                  21                 17.3725        1.238095
144864                  34                 20.7053        0.588235
144865                  27                 18.8885        0.629630
144866                   9                  8.8088       29.777778

[144867 rows x 24 columns]
```

1.6 Dataset - Column Profiling:

1. **data** - tells whether the data is testing or training data
2. **trip_creation_time** – Timestamp of trip creation
3. **route_schedule_uuid** – Unique ID for a particular route schedule
4. **route_type** – Transportation type

a. **Full Truck Load:** FTL shipments get to the destination sooner, as the truck is making no other pickups or drop-offs along the way

b. **Carting:** Handling system consisting of small vehicles (carts)

5. **trip_uuid** - Unique ID given to a particular trip (A trip may include different source and destination centers)
6. **source_center** - Source ID of trip origin
7. **source_name** - Source Name of trip origin
8. **destination_center** – Destination ID
9. **destination_name** – Destination Name
10. **od_start_time** – Trip start time
11. **od_end_time** – Trip end time
12. **start_scan_to_end_scan** – Time taken to deliver from source to destination
13. **is_cutoff** – Unknown field
14. **cutoff_factor** – Unknown field
15. **cutoff_timestamp** – Unknown field
16. **actual_distance_to_destination** – Distance in kms between source and destination warehouse
17. **actual_time** – Actual time taken to complete the delivery **(Cumulative)**
18. **osrm_time** – An open-source routing engine time calculator which computes the shortest path between points in a given map (Includes usual traffic, distance through major and minor roads) and gives the time **(Cumulative)**

19. **osrm_distance** – An open-source routing engine which computes the shortest path between points in a given map (Includes usual traffic, distance through major and minor roads) **(Cumulative)**
20. **factor** – Unknown field
21. **segment_actual_time** – This is a segment time. Time taken by the subset of the package delivery
22. **segment_osrm_time** – This is the OSRM segment time. Time taken by the subset of the package delivery
23. **segment_osrm_distance** – This is the OSRM distance. Distance covered by subset of the package delivery
24. **segment_factor** – Unknown field

1.7 Exploration and Data Cleaning

[ ]:
```
print("The number of rows:", df.shape[0])
print("The number of columns:", df.shape[1])
print("====="*10)
df.info()
```

```
The number of rows: 144867
The number of columns: 24
==========================================
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144867 entries, 0 to 144866
Data columns (total 24 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   data                            144867 non-null  object
 1   trip_creation_time              144867 non-null  object
 2   route_schedule_uuid             144867 non-null  object
 3   route_type                      144867 non-null  object
 4   trip_uuid                       144867 non-null  object
 5   source_center                   144867 non-null  object
 6   source_name                     144574 non-null  object
 7   destination_center              144867 non-null  object
 8   destination_name                144606 non-null  object
 9   od_start_time                   144867 non-null  object
 10  od_end_time                     144867 non-null  object
 11  start_scan_to_end_scan          144867 non-null  int64
 12  is_cutoff                       144867 non-null  bool
 13  cutoff_factor                   144867 non-null  int64
 14  cutoff_timestamp                144867 non-null  object
 15  actual_distance_to_destination  144867 non-null  float64
 16  actual_time                     144867 non-null  int64
 17  osrm_time                       144867 non-null  int64
```

```
 18  osrm_distance                   144867 non-null  float64
 19  factor                          144867 non-null  float64
 20  segment_actual_time             144867 non-null  int64
 21  segment_osrm_time               144867 non-null  int64
 22  segment_osrm_distance           144867 non-null  float64
 23  segment_factor                  144867 non-null  float64
dtypes: bool(1), float64(5), int64(6), object(12)
memory usage: 25.6+ MB
```

The dataset contains 144,867 rows and 24 columns. However, there are only 14,817 unique trip_uuid values. This confirms that the dataset is recorded at a segment level rather than a trip level, meaning that each trip is divided into multiple rows representing different segments of the journey.

On average, each trip consists of approximately 9–10 segment records. This indicates that aggregation will be required later to construct trip-level summaries for forecasting and business analysis.

[ ]:
```
print("Unique Trips:", df['trip_uuid'].nunique())
print("Total Rows:", len(df))
```

```
Unique Trips: 14817
Total Rows: 144867
```

The dataset contains 14,817 unique trips across 144,867 rows.

This confirms that the dataset is recorded at a segment level, meaning multiple rows correspond to a single trip.

On average:

144,867 / 14,817 9.77 segments per trip

This indicates that aggregation will be required later to construct trip-level summaries.

[ ]:
```
# Coverting necessary columns into their data types
datetime_cols =␣
 ↪['trip_creation_time','od_start_time','od_end_time','cutoff_timestamp']

for col in datetime_cols:
    df[col] = pd.to_datetime(df[col])
```

All time-related columns were initially stored as object data type. These were converted into pandas datetime format to enable time-based feature extraction and duration calculations. Converting these columns ensures accurate time computations and supports further analysis such as trip duration, monthly trends, and time-based aggregations.

[ ]:
```
df.duplicated().sum()
```

[ ]:
```
np.int64(0)
```

[ ]:
```
missing_percent = (df.isna().sum()/len(df))*100
missing_percent
```

[ ]:
```
data                              0.000000
trip_creation_time                0.000000
route_schedule_uuid               0.000000
route_type                        0.000000
trip_uuid                         0.000000
source_center                     0.000000
source_name                       0.202254
destination_center                0.000000
destination_name                  0.180165
od_start_time                     0.000000
od_end_time                       0.000000
start_scan_to_end_scan            0.000000
is_cutoff                         0.000000
cutoff_factor                     0.000000
cutoff_timestamp                  0.000000
actual_distance_to_destination    0.000000
actual_time                       0.000000
osrm_time                         0.000000
osrm_distance                     0.000000
factor                            0.000000
segment_actual_time               0.000000
segment_osrm_time                 0.000000
segment_osrm_distance             0.000000
segment_factor                    0.000000
dtype: float64
```

Missing values exist only in:

- source_name: 293 rows (0.20%)
- destination_name: 261 rows (0.18%)

Since missing values account for less than 0.5% of the dataset, they can be safely removed.

[ ]:
```
df = df.dropna(subset=['source_name','destination_name'])
```

After removing 551 rows (0.38% of total data), the dataset now contains:

[ ]:
```
print("Updated Rows:", len(df))
print("Updated Unique Trips:", df['trip_uuid'].nunique())
```

```
Updated Rows: 144316
Updated Unique Trips: 14787
```

[ ]:
```
(df['od_end_time'] < df['od_start_time']).sum()
(df['actual_time'] <= 0).sum()
df.duplicated().sum()
(df['actual_distance_to_destination'] < 0).sum()
```

[ ]: np.int64(0)

- No cases where od_end_time is earlier than od_start_time
- No negative or zero actual_time values
- No duplicate rows
- No negative distance values

This confirms structural consistency of the dataset.

[ ]:
```
df['data'].value_counts(normalize=True)*100
```

[ ]:
```
data
training    72.502009
test        27.497991
Name: proportion, dtype: float64
```

The data column tells us whether a particular record belongs to Training or Test dataset

This means the original dataset was likely prepared for a machine learning task, where:

- 72.5% of records are used to train a model
- 27.5% are reserved to test model performance

[ ]:
```
df['route_type'].value_counts(normalize=True)*100
```

[ ]:
```
route_type
FTL        68.690928
Carting    31.309072
Name: proportion, dtype: float64
```

- FTL (Full Truck Load): 68.69%
- Carting: 31.31%

Nearly two-thirds of shipments are handled via FTL. Since FTL shipments involve dedicated trucks, delivery performance may differ significantly from Carting shipments.

This distinction will be important during further analysis.

[ ]:
```
df.describe()
```

[ ]:
```
                  trip_creation_time                  od_start_time  \
count                          144316                         144316
mean   2018-09-22 13:05:09.445321216  2018-09-22 17:32:42.431788800
min              2018-09-12 00:00:17            2018-09-12 00:00:17
25%              2018-09-17 02:46:11  2018-09-17 07:37:35.249999872
50%              2018-09-22 03:36:19            2018-09-22 07:35:23
75%              2018-09-27 17:53:19            2018-09-27 22:01:31
max              2018-10-03 23:59:43            2018-10-06 04:27:23
```

```
std                                NaN                                NaN

                         od_end_time  start_scan_to_end_scan  cutoff_factor  \
count                         144316           144316.000000  144316.000000
mean   2018-09-23 09:36:54.055503104              963.697698     233.561345
min              2018-09-12 00:50:11               20.000000       9.000000
25%              2018-09-18 01:29:57              161.000000      22.000000
50%              2018-09-23 02:49:01              451.000000      66.000000
75%              2018-09-28 12:13:42             1645.000000     286.000000
max              2018-10-08 03:00:24             7898.000000    1927.000000
std                              NaN             1038.082976     345.245823

                    cutoff_timestamp  actual_distance_to_destination  \
count                         144316                    144316.000000
mean   2018-09-23 02:03:47.744546816                       234.708498
min              2018-09-12 00:02:10                         9.000045
25%    2018-09-17 18:30:36.750000128                        23.352027
50%              2018-09-22 19:27:22                        66.135322
75%       2018-09-28 04:56:21.500000                       286.919294
max              2018-10-06 23:44:12                      1927.447705
std                              NaN                       345.480571

         actual_time      osrm_time  osrm_distance         factor  \
count  144316.000000  144316.000000  144316.000000  144316.000000
mean      417.996237     214.437055     285.549785       2.120178
min         9.000000       6.000000       9.008200       0.144000
25%        51.000000      27.000000      29.896250       1.604545
50%       132.000000      64.000000      78.624400       1.857143
75%       516.000000     259.000000     346.305400       2.212280
max      4532.000000    1686.000000    2326.199100      77.387097
std       598.940065     308.448543     421.717826       1.717065

       segment_actual_time  segment_osrm_time  segment_osrm_distance  \
count        144316.000000      144316.000000          144316.000000
mean             36.175379          18.495697              22.818993
min            -244.000000           0.000000               0.000000
25%              20.000000          11.000000              12.053975
50%              28.000000          17.000000              23.508300
75%              40.000000          22.000000              27.813325
max            3051.000000        1611.000000            2191.403700
std              53.524298          14.774008              17.866367

       segment_factor
count   144316.000000
mean         2.218707
min        -23.444444
25%          1.347826
```

```
50%          1.684211
75%          2.250000
max        574.250000
std          4.854804
```

Key Observations

- Average actual delivery time: 417.99 minutes (~6.97 hours)
- Average OSRM estimated time: 214.44 minutes (~3.57 hours)

Actual time is nearly 95% higher than estimated time on average, suggesting systematic underestimation by OSRM.

- Average actual distance: 234.71 km
- Maximum actual_time: 4,532 minutes (~75.5 hours)

The wide range indicates presence of extreme outliers.

Preliminary Outlier Observations

From descriptive statistics:

- segment_actual_time minimum: -244 minutes (logically impossible)
- segment_factor minimum: -23.44
- segment_factor maximum: 574.25
- segment_actual_time maximum: 3,051 minutes

Negative time values are not possible in real-world logistics, indicating data anomalies. These extreme values will be handled later using IQR-based outlier detection.

[ ]:
```
(df['actual_time'] < df['osrm_time']).sum()
```

[ ]:
```
np.int64(2547)
```

[ ]:
```
(df['actual_time'] > df['osrm_time']).mean()
```

[ ]:
```
np.float64(0.9785332187699216)
```

Observation

- In 2,547 cases (1.76%), actual_time is less than osrm_time.
- In 97.85% of cases, actual_time exceeds osrm_time.

This suggests OSRM systematically underestimates real delivery time.

Possible reasons:

- Traffic congestion
- Loading/unloading delays
- Operational inefficiencies

- Route diversions

[ ]:
```
print(df['trip_creation_time'].min())
print(df['trip_creation_time'].max())
```

```
2018-09-12 00:00:17
2018-10-03 23:59:43
```

The dataset covers deliveries from **12 September 2018 to 3 October 2018**. This spans approximately 21 days of operations. While sufficient for operational analysis, the dataset does not capture long-term seasonal patterns.

[ ]:
```
df[['actual_time','actual_distance_to_destination']].corr()
```

[ ]:

	actual_time	actual_distance_to_destination
actual_time	1.000000	0.978683
actual_distance_to_destination	0.978683	1.000000

Observation

- There is an extremely strong positive correlation (97.87%) between delivery time and delivery distance.

Business Insight

This indicates that delivery time increases almost proportionally with distance. Distance is a dominant factor influencing delivery duration.

A correlation this high suggests:

- Distance is a strong predictor for delivery time.
- Forecasting models will heavily rely on distance-based features.
- Operational delays may scale with route length.

[ ]:
```
# Average Difference Between Actual and OSRM Time
df['time_difference'] = df['actual_time'] - df['osrm_time']
df['time_difference'].mean()
```

[ ]:
```
np.float64(203.55918262701294)
```

On average, actual delivery time exceeds OSRM estimated time by approximately 203 minutes (~3.4 hours).

Business Insight

This is a significant deviation.

It suggests:

- OSRM routing engine underestimates delivery time.
- Operational factors such as traffic congestion, loading delays, route diversions, and hub processing time are not fully captured in the route estimation model.

- There is a consistent positive bias in actual delivery time.

[ ]:
```
df[['actual_time','osrm_time','actual_distance_to_destination']].skew()
```

[ ]:
```
actual_time                       2.062382
osrm_time                         2.039250
actual_distance_to_destination    1.985217
dtype: float64
```

- The distributions are highly positively skewed (right-skewed).

The high skewness suggests that while most deliveries operate within expected time ranges, a small number of routes experience substantial delays.These extreme delays may be caused by:

- Long-haul interstate routes
- Traffic congestion
- Operational bottlenecks
- Multi-hub routing

[ ]:
```
# Dropping unknown fields
df.
 ↪drop(columns=['is_cutoff','cutoff_factor','cutoff_timestamp','factor','segment_factor'],␣
 ↪inplace=True)
df.columns
```

[ ]:
```
Index(['data', 'trip_creation_time', 'route_schedule_uuid', 'route_type',
       'trip_uuid', 'source_center', 'source_name', 'destination_center',
       'destination_name', 'od_start_time', 'od_end_time',
       'start_scan_to_end_scan', 'actual_distance_to_destination',
       'actual_time', 'osrm_time', 'osrm_distance', 'segment_actual_time',
       'segment_osrm_time', 'segment_osrm_distance', 'time_difference'],
      dtype='object')
```

Certain columns such as is_cutoff, cutoff_factor, and cutoff_timestamp were removed from the dataset as their business definitions were unclear and they were not required for the objectives of this case study.

These columns do not directly contribute to delivery time forecasting or segment aggregation. Removing them simplifies the dataset and reduces noise without affecting core delivery metrics.

1.8 Merge Rows

Here, in the given dataset, delivery details of one package is divided into several rows i.e. Each trip_uuid has multiple rows and Each row represents a segment of the journey. We need to merge rows properly to create segment-level summaries.

[ ]:
```

```

```
# Create a unique identifier using
df['segment_key'] = df['trip_uuid'] + '_' + df['source_center'] + '_' +␣
↪df['destination_center']
df[['trip_uuid','source_center','destination_center','segment_key']].head()
```

[ ]:
```
                 trip_uuid source_center destination_center  \
0  trip-153741093647649320  IND388121AAA       IND388620AAB
1  trip-153741093647649320  IND388121AAA       IND388620AAB
2  trip-153741093647649320  IND388121AAA       IND388620AAB
3  trip-153741093647649320  IND388121AAA       IND388620AAB
4  trip-153741093647649320  IND388121AAA       IND388620AAB

                                          segment_key
0  trip-153741093647649320_IND388121AAA_IND388620AAB
1  trip-153741093647649320_IND388121AAA_IND388620AAB
2  trip-153741093647649320_IND388121AAA_IND388620AAB
3  trip-153741093647649320_IND388121AAA_IND388620AAB
4  trip-153741093647649320_IND388121AAA_IND388620AAB
```

A new column segment_key was created using the combination of trip_uuid, source_center, and destination_center.

This uniquely identifies each delivery segment within a trip and allows grouping of rows belonging to the same segment.

[ ]:
```
df = df.sort_values(['segment_key','od_end_time'])
```

The dataset was sorted by segment_key and od_end_time to maintain chronological order of segment events. This is necessary before applying cumulative aggregation functions.

[ ]:
```
# merging rows
df['segment_actual_time_sum'] = df.
↪groupby('segment_key')['segment_actual_time'].cumsum()
df['segment_osrm_time_sum'] = df.groupby('segment_key')['segment_osrm_time'].
↪cumsum()
df['segment_osrm_distance_sum'] = df.
↪groupby('segment_key')['segment_osrm_distance'].cumsum()
# verifying the merged rows
df[['segment_key','segment_actual_time','segment_actual_time_sum','segment_osrm_time',
  ␣
↪'segment_osrm_time_sum','segment_osrm_distance','segment_osrm_distance_sum']].
↪head(10)
```

[ ]:
```
                                          segment_key  \
125002  trip-153671041653548748_IND209304AAA_IND000000ACB
125003  trip-153671041653548748_IND209304AAA_IND000000ACB
125004  trip-153671041653548748_IND209304AAA_IND000000ACB
125005  trip-153671041653548748_IND209304AAA_IND000000ACB
```

```
125006  trip-153671041653548748_IND209304AAA_IND000000ACB
125007  trip-153671041653548748_IND209304AAA_IND000000ACB
125008  trip-153671041653548748_IND209304AAA_IND000000ACB
125009  trip-153671041653548748_IND209304AAA_IND000000ACB
125010  trip-153671041653548748_IND209304AAA_IND000000ACB
125011  trip-153671041653548748_IND209304AAA_IND000000ACB

        segment_actual_time  segment_actual_time_sum  segment_osrm_time  \
125002                   50                       50                 33
125003                    9                       59                  8
125004                   34                       93                 22
125005                   25                      118                 17
125006                   20                      138                 14
125007                   38                      176                 81
125008                   41                      217                 78
125009                   50                      267                 28
125010                   56                      323                 19
125011                   28                      351                 16

        segment_osrm_time_sum  segment_osrm_distance  \
125002                     33                38.0620
125003                     41                10.9921
125004                     63                30.6308
125005                     80                25.0959
125006                     94                21.0389
125007                    175                79.9645
125008                    253               101.3602
125009                    281                40.8743
125010                    300                27.0127
125011                    316                23.6588

        segment_osrm_distance_sum
125002                    38.0620
125003                    49.0541
125004                    79.6849
125005                   104.7808
125006                   125.8197
125007                   205.7842
125008                   307.1444
125009                   348.0187
125010                   375.0314
125011                   398.6902
```

Cumulative sums were computed for segment_actual_time, segment_osrm_time, and segment_osrm_distance, lastly grouped by segment_key.

These new columns represent the running total time and distance within each segment, enabling proper aggregation of multiple rows belonging to the same delivery segment.

[ ]:
```
creating_dict_segment = {

    # categorical / constant columns
    'data': 'first',
    'trip_creation_time': 'first',
    'route_schedule_uuid': 'first',
    'route_type': 'first',
    'trip_uuid': 'first',
    'source_center': 'first',
    'source_name': 'first',
    'destination_center': 'first',
    'destination_name': 'first',
    'od_start_time': 'first',

    # final values
    'od_end_time': 'last',
    'actual_distance_to_destination': 'last',
    'actual_time': 'last',
    'osrm_time': 'last',
    'osrm_distance': 'last',
    'time_difference': 'last',

    # cumulative segment totals
    'segment_actual_time_sum': 'last',
    'segment_osrm_time_sum': 'last',
    'segment_osrm_distance_sum': 'last'
}
```

[ ]:
```
segment = df.groupby('segment_key').agg(creating_dict_segment).reset_index()
```

[ ]:
```
segment = segment.sort_values(['segment_key','od_end_time'])
```

[ ]:
```
print("Original rows:", df.shape)
print("Segment rows:", segment.shape)

segment.head()
```

```
Original rows: (144316, 24)
Segment rows: (26222, 20)
```

[ ]:
```
                                          segment_key      data  \
0  trip-153671041653548748_IND209304AAA_IND000000ACB  training
1  trip-153671041653548748_IND462022AAA_IND209304AAA  training
2  trip-153671042288605164_IND561203AAB_IND562101AAA  training
3  trip-153671042288605164_IND572101AAA_IND561203AAB  training
4  trip-153671043369099517_IND000000ACB_IND160002AAC  training
```

```
   trip_creation_time                              route_schedule_uuid  \
0 2018-09-12 00:00:17  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
1 2018-09-12 00:00:17  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
2 2018-09-12 00:00:23  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…
3 2018-09-12 00:00:23  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…
4 2018-09-12 00:00:34  thanos::sroute:de5e208e-7641-45e6-8100-4d9fb1e…

  route_type                trip_uuid source_center  \
0        FTL  trip-153671041653548748  IND209304AAA
1        FTL  trip-153671041653548748  IND462022AAA
2    Carting  trip-153671042288605164  IND561203AAB
3    Carting  trip-153671042288605164  IND572101AAA
4        FTL  trip-153671043369099517  IND000000ACB

                        source_name destination_center  \
0  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
1  Bhopal_Trnsport_H (Madhya Pradesh)       IND209304AAA
2   Doddablpur_ChikaDPP_D (Karnataka)       IND562101AAA
3       Tumkur_Veersagr_I (Karnataka)       IND561203AAB
4       Gurgaon_Bilaspur_HB (Haryana)       IND160002AAC

                     destination_name       od_start_time         od_end_time  \
0       Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:47 2018-09-13 13:40:23
1  Kanpur_Central_H_6 (Uttar Pradesh) 2018-09-12 00:00:17 2018-09-12 16:39:47
2   Chikblapur_ShntiSgr_D (Karnataka) 2018-09-12 02:03:10 2018-09-12 03:02:00
3   Doddablpur_ChikaDPP_D (Karnataka) 2018-09-12 00:00:23 2018-09-12 02:03:10
4      Chandigarh_Mehmdpur_H (Punjab) 2018-09-14 03:40:17 2018-09-14 17:34:55

   actual_distance_to_destination  actual_time  osrm_time  osrm_distance  \
0                      383.759164          732        329       446.5496
1                      440.973689          830        388       544.8027
2                       24.644021           47         26        28.1994
3                       48.542890           96         42        56.9116
4                      237.439610          611        212       281.2109

   time_difference  segment_actual_time_sum  segment_osrm_time_sum  \
0              403                      728                    534
1              442                      820                    474
2               21                       46                     26
3               54                       95                     39
4              399                      608                    231

   segment_osrm_distance_sum
0                   670.6205
1                   649.8528
2                    28.1995
3                    55.9899
```

```
4                    317.7408
```

After generating cumulative segment metrics, rows belonging to the same delivery segment were aggregated using segment_key. A dictionary named create_segment_dict was created to specify aggregation logic for each column.

Categorical fields such as route type, source, and destination were preserved using the first value since they remain constant within a segment. Time-based cumulative fields such as actual_time, osrm_time, and actual_distance_to_destination were captured using the last value to obtain the final state of the segment.

Additionally, the cumulative segment metrics (segment_actual_time_sum, segment_osrm_time_sum, and segment_osrm_distance_sum) were aggregated using last, as the final row already represents the total accumulated value.

This transformation reduced multiple rows representing the same segment into a single consolidated record per segment.

[ ]:
```
# Create Trip Aggregation Dictionary
create_trip_dict = {

    'data':'first',
    'trip_creation_time':'first',
    'route_schedule_uuid':'first',
    'route_type':'first',

    'source_center':'first',
    'source_name':'first',

    'destination_center':'last',
    'destination_name':'last',

    'od_start_time':'first',
    'od_end_time':'last',

    'actual_distance_to_destination':'last',
    'actual_time':'last',

    'osrm_time':'last',
    'osrm_distance':'last',

    'segment_actual_time_sum':'sum',
    'segment_osrm_time_sum':'sum',
    'segment_osrm_distance_sum':'sum'
}
```

[ ]:
```
trip = segment.groupby('trip_uuid').agg(create_trip_dict).reset_index()
print("Segment rows:", segment.shape)
print("Trip rows:", trip.shape)
```

```
trip.head()
```

```
Segment rows: (26222, 20)
Trip rows: (14787, 18)
```

[ ]:

```
                 trip_uuid      data  trip_creation_time  \
0  trip-153671041653548748  training 2018-09-12 00:00:17
1  trip-153671042288605164  training 2018-09-12 00:00:23
2  trip-153671043369099517  training 2018-09-12 00:00:34
3  trip-153671046011330457  training 2018-09-12 00:01:00
4  trip-153671052974046625  training 2018-09-12 00:02:10

                                 route_schedule_uuid route_type source_center  \
0  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…        FTL  IND209304AAA
1  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…    Carting  IND561203AAB
2  thanos::sroute:de5e208e-7641-45e6-8100-4d9fb1e…        FTL  IND000000ACB
3  thanos::sroute:f0176492-a679-4597-8332-bbd1c7f…    Carting  IND400072AAB
4  thanos::sroute:d9f07b12-65e0-4f3b-bec8-df06134…        FTL  IND583101AAA

                         source_name destination_center  \
0  Kanpur_Central_H_6 (Uttar Pradesh)       IND209304AAA
1   Doddablpur_ChikaDPP_D (Karnataka)       IND561203AAB
2       Gurgaon_Bilaspur_HB (Haryana)       IND000000ACB
3            Mumbai Hub (Maharashtra)       IND401104AAA
4              Bellary_Dc (Karnataka)       IND583119AAA

                     destination_name       od_start_time         od_end_time  \
0  Kanpur_Central_H_6 (Uttar Pradesh) 2018-09-12 16:39:47 2018-09-12 16:39:47
1   Doddablpur_ChikaDPP_D (Karnataka) 2018-09-12 02:03:10 2018-09-12 02:03:10
2       Gurgaon_Bilaspur_HB (Haryana) 2018-09-14 03:40:17 2018-09-14 03:40:17
3      Mumbai_MiraRd_IP (Maharashtra) 2018-09-12 00:01:00 2018-09-12 01:41:30
4       Sandur_WrdN1DPP_D (Karnataka) 2018-09-12 00:02:10 2018-09-12 03:54:43

   actual_distance_to_destination  actual_time  osrm_time  osrm_distance  \
0                      440.973689          830        388       544.8027
1                       48.542890           96         42        56.9116
2                     1689.964663         2736       1528      2072.8556
3                       17.175274           59         15        19.6800
4                       26.600536           63         27        29.5696

   segment_actual_time_sum  segment_osrm_time_sum  segment_osrm_distance_sum
0                     1548                   1008                  1320.4733
1                      141                     65                    84.1894
2                     3308                   1941                  2545.2678
3                       59                     16                    19.8766
4                      340                    115                   146.7919
```

[ ]:
```
trip = trip.sort_values(['trip_creation_time'])
```

After segment-level aggregation, each trip still consisted of multiple segments representing intermediate transportation points. Therefore, the dataset was further aggregated at the trip level using trip_uuid.

Columns that remain constant across the trip were aggregated using first, while the final values of cumulative metrics such as actual_time, osrm_time, and actual_distance_to_destination were obtained using last.

Segment-level metrics such as segment_actual_time_sum, segment_osrm_time_sum, and segment_osrm_distance_sum were aggregated using sum to compute the total travel time and distance across all segments within a trip.

This transformation produced a final dataset where each row represents a complete trip from origin to final destination.

[ ]:
```
print(trip[['actual_time','segment_actual_time_sum']].head())
```

```
   actual_time  segment_actual_time_sum
0          830                     1548
1           96                      141
2         2736                     3308
3           59                       59
4           63                      340
```

After trip-level aggregation, actual_time and segment_actual_time_sum were compared to validate aggregation logic.

It was observed that segment_actual_time_sum is generally greater than actual_time. This is expected because segment_actual_time_sum represents the sum of operational time across individual delivery segments, while actual_time represents the cumulative trip duration. Segment-level operations may include additional activities such as loading, scanning, or handling delays that do not strictly extend the overall trip timeline.

Thus, the observed difference confirms that the aggregation logic has been applied correctly.

1.9 Feature Engineering - Extracting Features from Given Dataset

1.9.1 Calculate Trip Duration

[ ]:
```
# Create Trip Duration Feature
trip['trip_duration_hours'] = (trip['od_end_time'] - trip['od_start_time']).dt.
 ↪total_seconds() / 3600
trip = trip[trip['trip_duration_hours'] >= 0]
```

We compute trip duration.

1.9.2 Extract Time Features

[ ]:
```
# Extract Time Feature
trip['trip_month'] = trip['trip_creation_time'].dt.month
trip['trip_day'] = trip['trip_creation_time'].dt.day
trip['trip_weekday'] = trip['trip_creation_time'].dt.dayofweek
trip['trip_hour'] = trip['trip_creation_time'].dt.hour
```

1.9.3 Extract location

[ ]:
```
trip['source_location'] = trip['source_name'].str.split('(').str[0]
trip['destination_location'] = trip['destination_name'].str.split('(').str[0]
trip[['source_location', 'destination_location']].head(2)
```

[ ]:
```
          source_location    destination_location
0     Kanpur_Central_H_6      Kanpur_Central_H_6
1  Doddablpur_ChikaDPP_D   Doddablpur_ChikaDPP_D
```

1.9.4 Extract City

[ ]:
```
trip['source_city'] = trip['source_location'].str.split('_').str[0]
trip['destination_city'] = trip['destination_location'].str.split('_').str[0]
```

1.9.5 Extract State

[ ]:
```
trip['source_state'] = trip['source_name'].str.split('(').str[1].str.rstrip(')')
trip['destination_state'] = trip['destination_name'].str.split('(').str[1].str.
 ↪rstrip(')')
```

1.9.6 Calculate Speed

[ ]:
```
trip['avg_speed_kmph'] = trip['actual_distance_to_destination'] /␣
 ↪(trip['actual_time']/60)
```

1.9.7 Create OSRM Prediction Error

[ ]:
```
trip['time_error'] = trip['actual_time'] - trip['osrm_time']
```

1.9.8 Drop Unnecessary column

[ ]:
```
trip.drop(columns=['source_location','destination_location', 'source_name',␣
 ↪'destination_name'], inplace=True)
```

Temporary columns source_location and destination_location were created during the feature engineering process to extract city and state information from the location fields. After successfully

extracting source_city, destination_city, source_state, and destination_state, these intermediate columns were removed to simplify the dataset structure and avoid redundancy.

[ ]:
```python
# Checking Columns and Dataset after droping columns
print(trip.shape)
print('======' * 10)
trip.info()
```

```
(13896, 27)
============================================================
<class 'pandas.core.frame.DataFrame'>
Index: 13896 entries, 0 to 14786
Data columns (total 27 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   trip_uuid                       13896 non-null  object
 1   data                            13896 non-null  object
 2   trip_creation_time              13896 non-null  datetime64[ns]
 3   route_schedule_uuid             13896 non-null  object
 4   route_type                      13896 non-null  object
 5   source_center                   13896 non-null  object
 6   destination_center              13896 non-null  object
 7   od_start_time                   13896 non-null  datetime64[ns]
 8   od_end_time                     13896 non-null  datetime64[ns]
 9   actual_distance_to_destination  13896 non-null  float64
 10  actual_time                     13896 non-null  int64
 11  osrm_time                       13896 non-null  int64
 12  osrm_distance                   13896 non-null  float64
 13  segment_actual_time_sum         13896 non-null  int64
 14  segment_osrm_time_sum           13896 non-null  int64
 15  segment_osrm_distance_sum       13896 non-null  float64
 16  trip_duration_hours             13896 non-null  float64
 17  trip_month                      13896 non-null  int32
 18  trip_day                        13896 non-null  int32
 19  trip_weekday                    13896 non-null  int32
 20  trip_hour                       13896 non-null  int32
 21  source_city                     13896 non-null  object
 22  destination_city                13896 non-null  object
 23  source_state                    13896 non-null  object
 24  destination_state               13896 non-null  object
 25  avg_speed_kmph                  13896 non-null  float64
 26  time_error                      13896 non-null  int64
dtypes: datetime64[ns](3), float64(5), int32(4), int64(5), object(10)
memory usage: 2.8+ MB
```

1.9.9 Summary Statistics for Trip Dataset

[ ]:
```
trip.describe().T
```

[ ]:
```
                                  count                           mean  \
trip_creation_time                13896  2018-09-22 11:55:13.060304896
od_start_time                     13896  2018-09-22 13:43:48.160118016
od_end_time                       13896  2018-09-22 20:28:51.724596992
actual_distance_to_destination  13896.0                     106.489047
actual_time                     13896.0                     233.641048
osrm_time                       13896.0                     103.891192
osrm_distance                   13896.0                     132.373359
segment_actual_time_sum         13896.0                     346.004102
segment_osrm_time_sum           13896.0                     176.625288
segment_osrm_distance_sum       13896.0                     218.426796
trip_duration_hours             13896.0                        6.75099
trip_month                      13896.0                       9.119675
trip_day                        13896.0                      18.368164
trip_weekday                    13896.0                       2.925302
trip_hour                       13896.0                       12.41314
avg_speed_kmph                  13896.0                      25.916265
time_error                      13896.0                     129.749856

                                                   min  \
trip_creation_time              2018-09-12 00:00:17
od_start_time                   2018-09-12 00:01:00
od_end_time                     2018-09-12 00:50:11
actual_distance_to_destination             9.002461
actual_time                                     9.0
osrm_time                                       6.0
osrm_distance                                 9.0729
segment_actual_time_sum                         9.0
segment_osrm_time_sum                           6.0
segment_osrm_distance_sum                     9.0729
trip_duration_hours                             0.0
trip_month                                      9.0
trip_day                                        1.0
trip_weekday                                    0.0
trip_hour                                       0.0
avg_speed_kmph                             0.434947
time_error                                    -45.0

                                                         25%  \
trip_creation_time                 2018-09-17 01:56:02.500000
od_start_time                   2018-09-17 04:12:26.249999872
od_end_time                     2018-09-17 10:16:06.249999872
actual_distance_to_destination                      20.108492
```

```
actual_time                                          52.0
osrm_time                                            23.0
osrm_distance                                     26.1123
segment_actual_time_sum                              64.0
segment_osrm_time_sum                                30.0
segment_osrm_distance_sum                        31.352725
trip_duration_hours                               1.865556
trip_month                                            9.0
trip_day                                             14.0
trip_weekday                                          1.0
trip_hour                                             4.0
avg_speed_kmph                                   18.286862
time_error                                           23.0

                                                      50%  \
trip_creation_time              2018-09-22 03:26:29.500000
od_start_time                          2018-09-22 04:51:43
od_end_time                            2018-09-22 12:20:27
actual_distance_to_destination                   32.852955
actual_time                                           87.0
osrm_time                                             38.0
osrm_distance                                      41.1468
segment_actual_time_sum                              136.0
segment_osrm_time_sum                                 62.0
segment_osrm_distance_sum                          65.6092
trip_duration_hours                               3.515139
trip_month                                             9.0
trip_day                                              19.0
trip_weekday                                           3.0
trip_hour                                             14.0
avg_speed_kmph                                   25.846486
time_error                                            46.0

                                                       75%  \
trip_creation_time              2018-09-27 18:51:38.750000128
od_start_time                   2018-09-27 19:53:20.249999872
od_end_time                     2018-09-28 02:15:43.750000128
actual_distance_to_destination                     66.243612
actual_time                                           184.0
osrm_time                                              74.0
osrm_distance                                     86.187525
segment_actual_time_sum                               349.0
segment_osrm_time_sum                                 176.0
segment_osrm_distance_sum                          204.5887
trip_duration_hours                                7.072639
trip_month                                             9.0
trip_day                                              25.0
```

```
trip_weekday                                          5.0
trip_hour                                            20.0
avg_speed_kmph                                  33.410902
time_error                                         109.25

                                                   max         std
trip_creation_time                 2018-10-03 23:59:43         NaN
od_start_time                      2018-10-06 04:27:23         NaN
od_end_time                        2018-10-08 03:00:24         NaN
actual_distance_to_destination            1927.447705  248.333932
actual_time                                    4532.0  455.362551
osrm_time                                      1686.0  219.503269
osrm_distance                               2326.1991  301.557762
segment_actual_time_sum                        6230.0  561.772403
segment_osrm_time_sum                          2564.0  316.600496
segment_osrm_distance_sum                   3523.6324  419.967544
trip_duration_hours                          131.6425    9.562075
trip_month                                       10.0    0.324592
trip_day                                         30.0    7.873699
trip_weekday                                      6.0    1.928004
trip_hour                                        23.0    7.948861
avg_speed_kmph                             124.077312   10.581474
time_error                                     3137.0  252.920048
```

Summary Statistics – Observations The summary statistics provide an overview of the key trip-level variables such as delivery time, distance, and routing estimates.

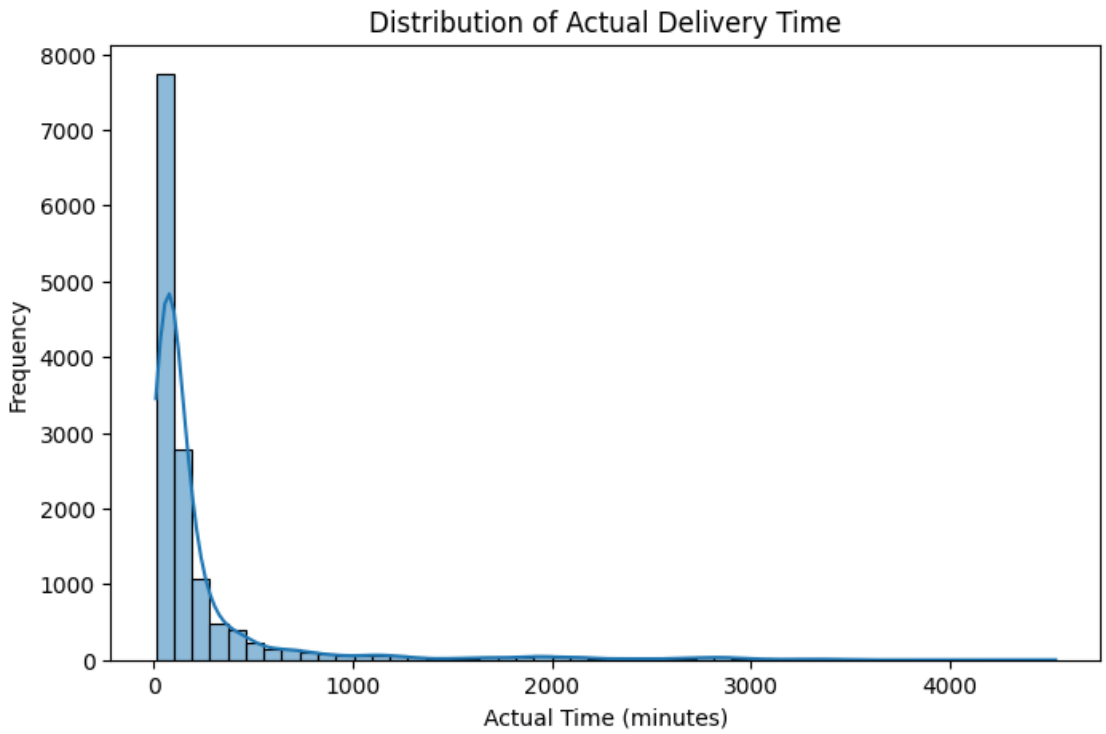
- The average delivery distance is 106 km, while the median is 33 km, indicating that most deliveries are short-distance, with a few long-haul trips increasing the average.
- The average actual delivery time is 234 minutes (~3.9 hours), whereas the median is 87 minutes (approx 1.5 hours), suggesting a right-skewed distribution where a few trips take significantly longer than usual.
- The OSRM estimated time (104 minutes on average) is much lower than the actual delivery time (234 minutes), indicating that route estimates tend to underestimate real-world delivery duration by about 130 minutes on average.
- The average delivery speed is around 26 km/h, which reflects typical logistics movement considering traffic and operational delays.
- The maximum delivery distance of 1,927 km and maximum delivery time of 4,532 minutes (~75.5 hours) indicate the presence of extreme outliers or inter-city/long-haul trips.
- Negative values in time_error (as low as -45) indicate trips that were completed faster than OSRM estimated, while large positive values (up to 3,137) show significant delays.
- Some extreme values such as very large delivery times and negative trip duration indicate potential outliers or data anomalies that will be addressed in the outlier detection stage.

1.9.10 Univariate Analysis

Distribution of Actual Delivery Time

[ ]:
```
plt.figure(figsize=(8,5))
sns.histplot(trip['actual_time'], bins=50, kde=True)

plt.title("Distribution of Actual Delivery Time")
plt.xlabel("Actual Time (minutes)")
plt.ylabel("Frequency")
plt.show()
```

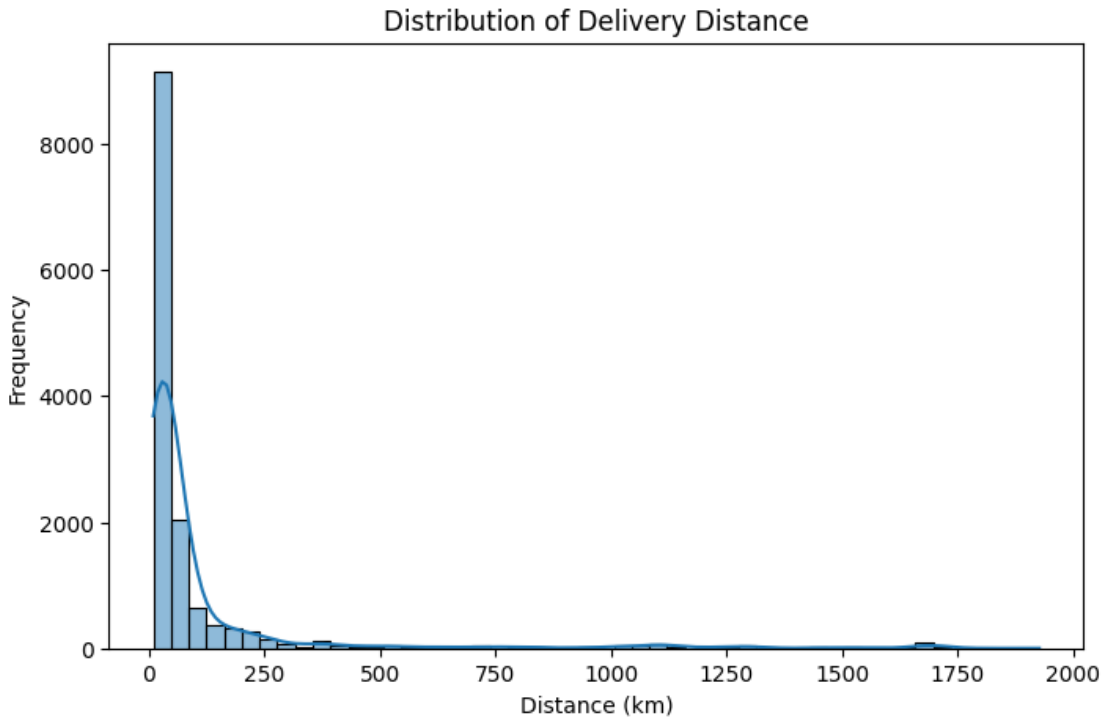


- Observation - The delivery time distribution is highly right-skewed. Most deliveries are completed within 0–300 minutes, while a small number of trips take significantly longer, extending beyond 2000 minutes.
- Insight - This pattern indicates that the majority of deliveries operate within normal operational timelines, but a small fraction of trips experience extreme delays. These delays could be caused by long-distance routes, traffic congestion, or operational bottlenecks at logistics hubs.

Distribution of Delhivery Distance

[ ]:
```
plt.figure(figsize=(8,5))
sns.histplot(trip['actual_distance_to_destination'], bins=50, kde=True)
```

```
plt.title("Distribution of Delivery Distance")
plt.xlabel("Distance (km)")
plt.ylabel("Frequency")
plt.show()
```

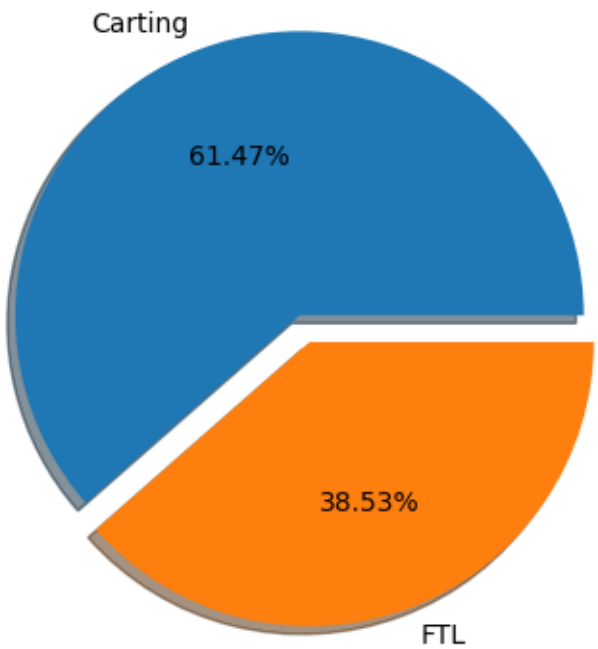


- Observation:- The distribution of delivery distance is also right-skewed. Most deliveries occur within short distances (below ~100 km), while a small number of long-haul deliveries extend up to 1900 km.
- Business Insight:- Delhivery's operations appear to be dominated by short-distance regional deliveries, with fewer long-distance interstate shipments. Long-haul deliveries likely correspond to inter-city or cross-state transportation routes.

Route Type Distribution

[ ]:
```
plt.pie(data=trip,x=trip['route_type'].
↪value_counts(),shadow=True,labels=['Carting','FTL'],explode=(0,0.
↪1),autopct='%0.2f%%')
plt.plot()
```

[ ]: []



- Observation :- Carting accounts for approximately 61.47% of trips, while FTL accounts for 38.53%.
- Business Insight :- Carting dominates the logistics operations, suggesting that multi-stop transportation using smaller vehicles is more common than full truckload shipments. This reflects the nature of e-commerce logistics where multiple packages are consolidated and distributed across several locations.

Destination State with Delhivery Count

[ ]:
```
x = trip.groupby('destination_state')['trip_uuid'].count().
↪sort_values(ascending=False).to_frame().reset_index()
x
```

[ ]:

	destination_state	trip_uuid
0	Maharashtra	2448
1	Karnataka	2243
2	Haryana	1616
3	Tamil Nadu	1034
4	Uttar Pradesh	731
5	Telangana	686
6	West Bengal	666
7	Delhi	657
8	Gujarat	622

```
9                   Punjab        558
10                Rajasthan        503
11           Andhra Pradesh        436
12                    Bihar        328
13           Madhya Pradesh        312
14                    Assam        226
15                   Kerala        206
16                Jharkhand        152
17                   Orissa        119
18              Uttarakhand        111
19                      Goa         52
20             Chhattisgarh         43
21         Himachal Pradesh         38
22               Chandigarh         30
23        Arunachal Pradesh         25
24          Jammu & Kashmir         20
25  Dadra and Nagar Haveli         17
26                Meghalaya          8
27                  Mizoram          6
28              Daman & Diu          1
29                 Nagaland          1
30                  Tripura          1
```

The highest number of deliveries are directed to:

- Maharashtra – 2448 trips
- Karnataka – 2243 trips

Business Insight

These states represent major logistics demand hubs. Maharashtra and Karnataka host large metropolitan areas such as Mumbai and Bengaluru, which are major e-commerce consumption centers.

Source State with Delhivery Count

[ ]:
```
y=trip.groupby('source_state')['trip_uuid'].count().
 ↪sort_values(ascending=False).to_frame().reset_index()
y
```

[ ]:
```
               source_state  trip_uuid
0               Maharashtra       2600
1                 Karnataka       2106
2                   Haryana       1793
3                 Tamil Nadu        989
4                     Delhi        725
5                 Telangana        687
6             Uttar Pradesh        681
7               West Bengal        634
```

```
8                  Gujarat        626
9                Rajasthan        473
10                  Punjab        467
11           Andhra Pradesh        429
12                    Bihar        316
13           Madhya Pradesh        279
14                    Assam        262
15                   Kerala        225
16                Jharkhand        131
17                   Orissa        107
18              Uttarakhand        103
19               Chandigarh         67
20                      Goa         65
21             Chhattisgarh         43
22         Himachal Pradesh         31
23          Jammu & Kashmir         17
24  Dadra and Nagar Haveli         15
25              Pondicherry         12
26                 Nagaland          5
27        Arunachal Pradesh          4
28                  Mizoram          4
```

Observation

The highest shipment origins are:

- Maharashtra – 2600 trips
- Karnataka – 2106 trips
- Haryana – 1793 trips

Business Insight

These states appear to be major logistics supply hubs, likely containing large warehouses or fulfillment centers from where shipments are dispatched to other regions.

Analaysing through destination city, Which place have more delivery ?

[ ]:
```
z=trip.groupby('destination_city')['trip_uuid'].count().
↪sort_values(ascending=False).head(11).to_frame().reset_index()
z
```

[ ]:
```
   destination_city  trip_uuid
0         Bengaluru       1088
1            Mumbai        913
2           Gurgaon        877
3             Delhi        554
4         Bangalore        538
5         Hyderabad        489
6          Bhiwandi        434
```

```
7           Chennai        406
8           Sonipat        322
9              Pune        313
10       Chandigarh        303
```

Observation

The cities receiving the highest number of deliveries include:

- Bengaluru
- Mumbai
- Gurgaon

Business Insight

These cities are major urban demand centers, indicating that delivery volume is concentrated in highly populated metropolitan regions where e-commerce activity is high.

Analaysing through source city, Which place have more delivery ?

[ ]:
```
w=trip.groupby('source_city')['trip_uuid'].count().sort_values(ascending=False).
↪head(11).to_frame().reset_index()
w
```

[ ]:
```
    source_city  trip_uuid
0       Gurgaon       1128
1     Bengaluru       1052
2      Bhiwandi        697
3     Bangalore        648
4        Mumbai        601
5         Delhi        548
6     Hyderabad        507
7          Pune        467
8       Chennai        334
9   Mumbai Hub         300
10      Kolkata        281
```

Observation

The most frequent shipment origins include:

- Gurgaon
- Bengaluru
- Bhiwandi

Business Insight

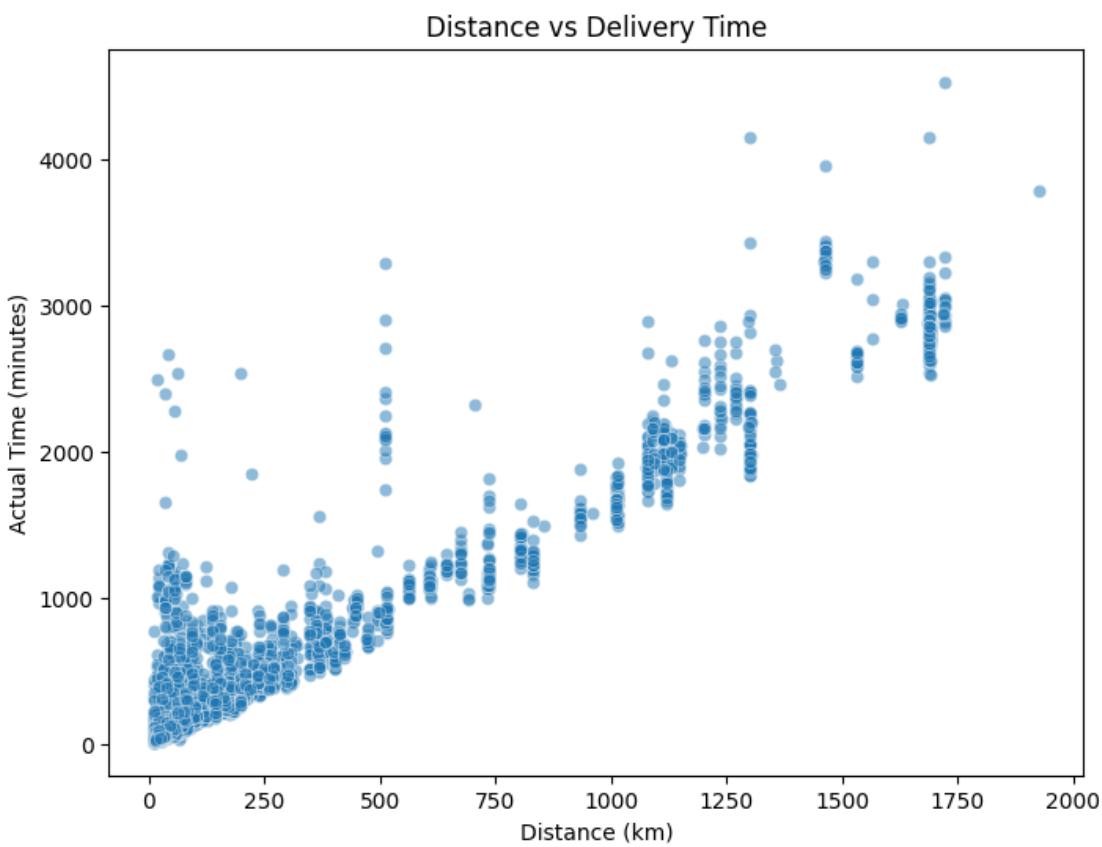
Cities such as Gurgaon and Bhiwandi are well-known logistics hubs in India, suggesting that many shipments originate from centralized warehouses located near these cities.

1.9.11 Bivariate Analysis

Distance vs Delivery Time

[ ]:
```
plt.figure(figsize=(8,6))
sns.
 ↪scatterplot(x='actual_distance_to_destination',y='actual_time',data=trip,alpha=0.
 ↪5)

plt.title("Distance vs Delivery Time")
plt.xlabel("Distance (km)")
plt.ylabel("Actual Time (minutes)")
plt.show()
```

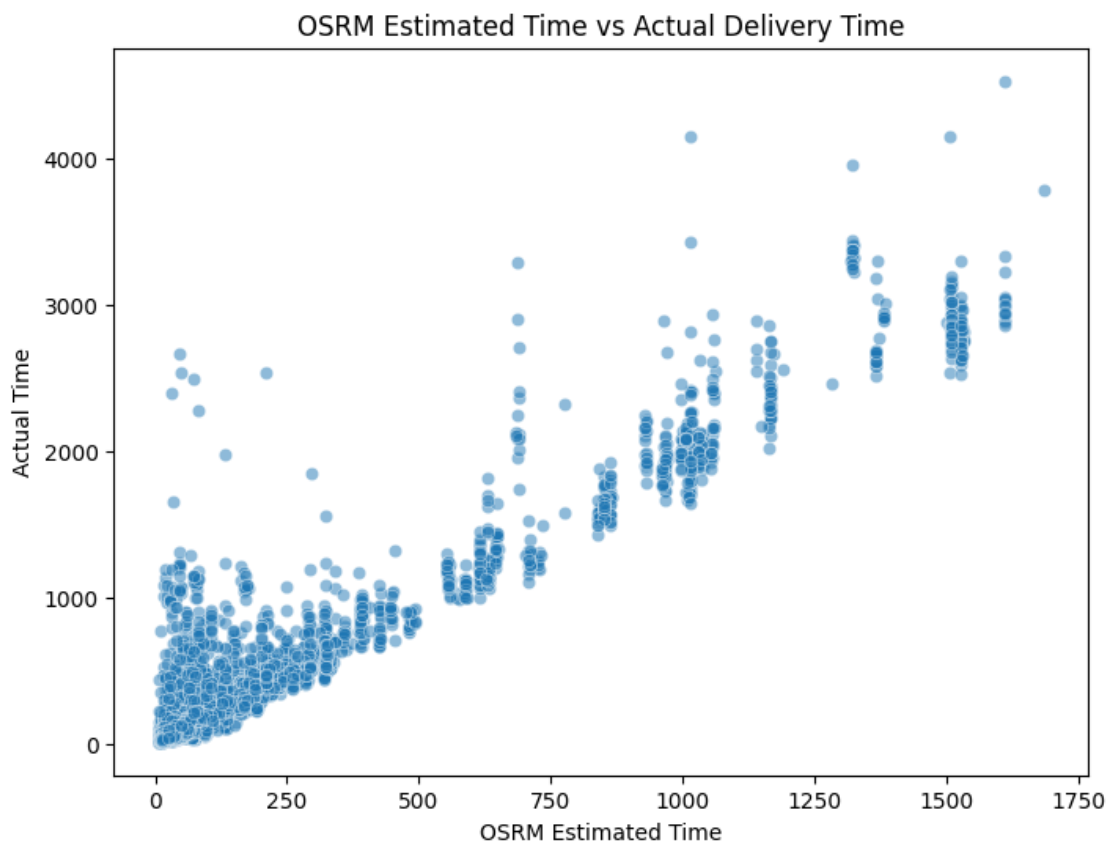


- Observation :- There is a strong positive relationship between delivery distance and delivery time. As the delivery distance increases, the actual delivery time also increases.
- Business Insight :- Distance is clearly one of the most important drivers of delivery time. This suggests that distance-based features will play a significant role in predicting delivery duration.

OSRM Time vs Actual Time

[ ]:
```
plt.figure(figsize=(8,6))
sns.scatterplot(x='osrm_time',y='actual_time',data=trip,alpha=0.5)

plt.title("OSRM Estimated Time vs Actual Delivery Time")
plt.xlabel("OSRM Estimated Time")
plt.ylabel("Actual Time")
plt.show()
```



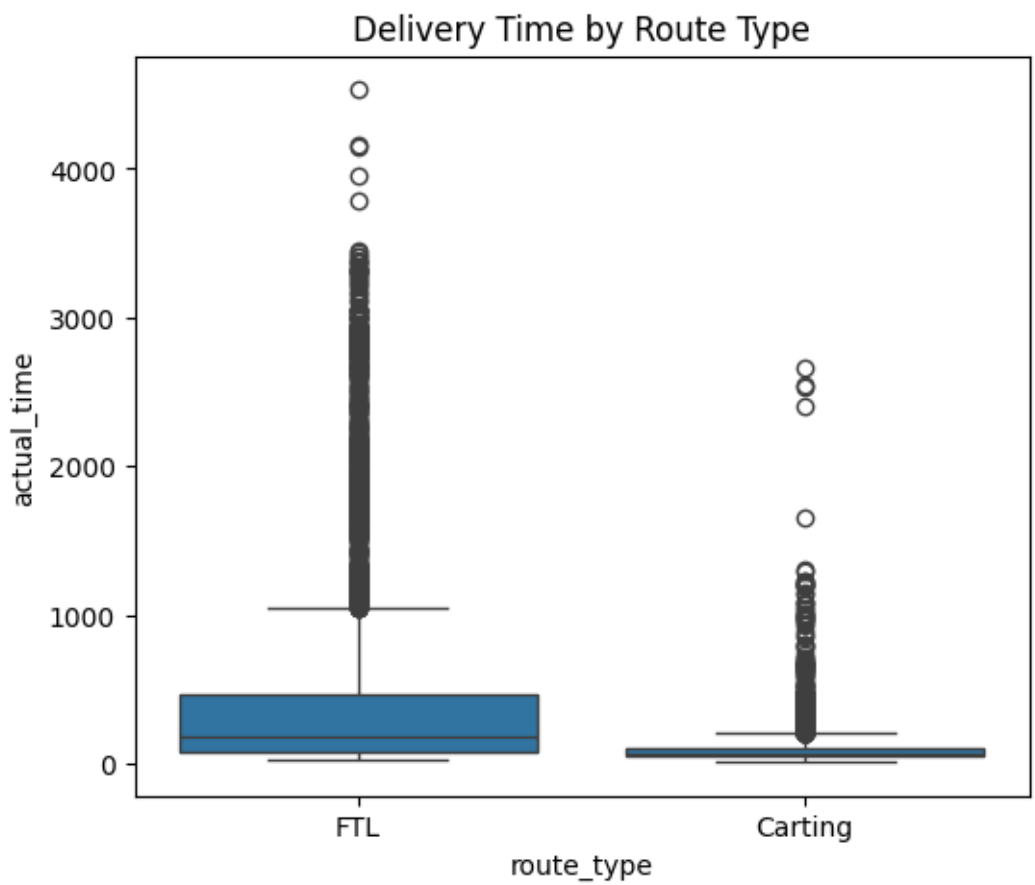
- Observation: The scatter plot shows that actual delivery time is usually higher than the OSRM estimated time.
- Business Insight: This indicates that OSRM route estimates consistently underestimate real delivery durations. The routing engine likely does not account for operational delays such as loading/unloading, traffic congestion, or intermediate hub processing.

Delivery Time by Route Type

[ ]:
```
plt.figure(figsize=(6,5))
sns.boxplot(x='route_type', y='actual_time', data=trip)

plt.title("Delivery Time by Route Type")
```

```
plt.show()
```



- Observation: FTL deliveries exhibit higher variability and longer delivery times compared to Carting shipments.
- Business Insight: FTL shipments typically involve long-distance routes, which explains the wider spread of delivery times. Carting shipments are generally shorter, local deliveries, leading to more consistent delivery durations.

1.9.12 Multivariate Analysis

Correlation Heatmap

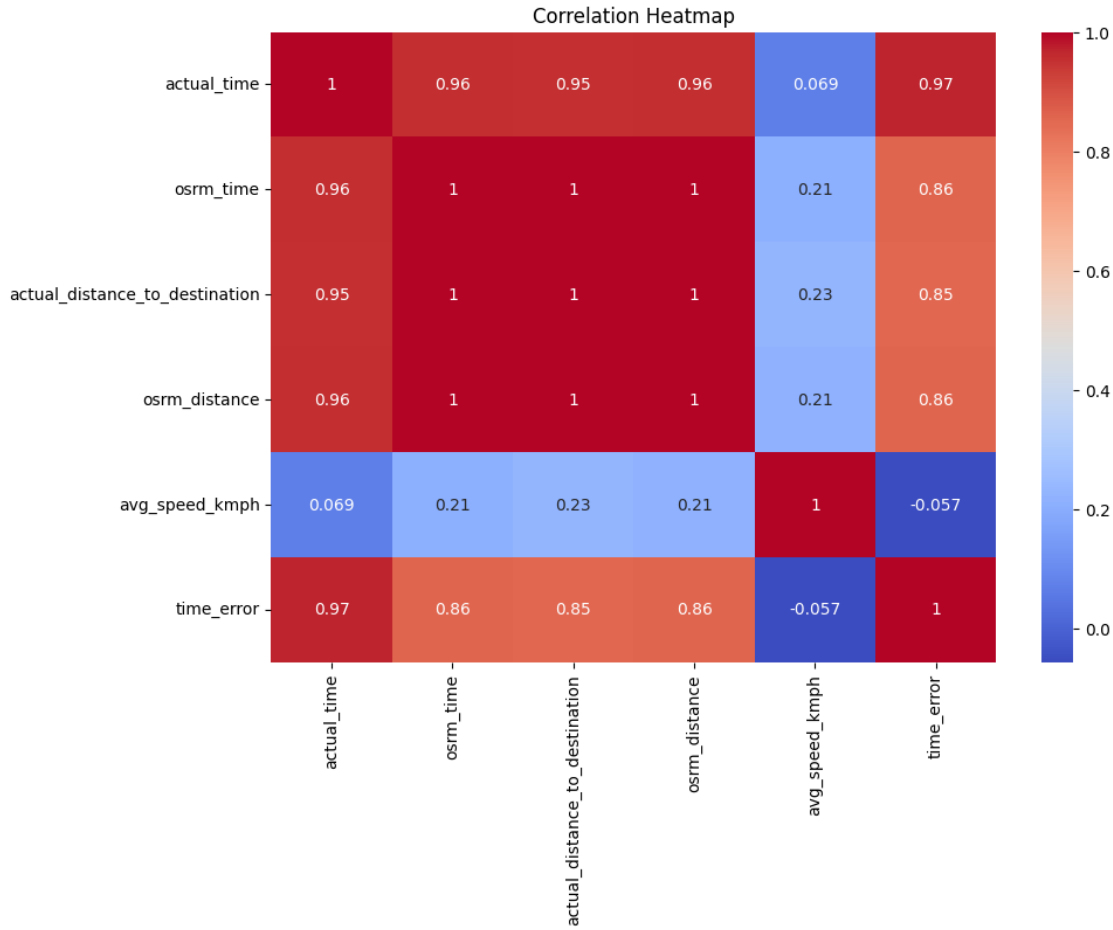
[ ]:
```
plt.figure(figsize=(10,7))

corr = trip[['actual_time','osrm_time','actual_distance_to_destination',
             'osrm_distance','avg_speed_kmph','time_error']].corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")
```

```
plt.show()
```



Observation

Strong positive correlations are observed between:

- Actual time and OSRM time (0.96)
- Actual time and actual distance (0.95)
- OSRM time and OSRM distance (~1.00)

Business Insight

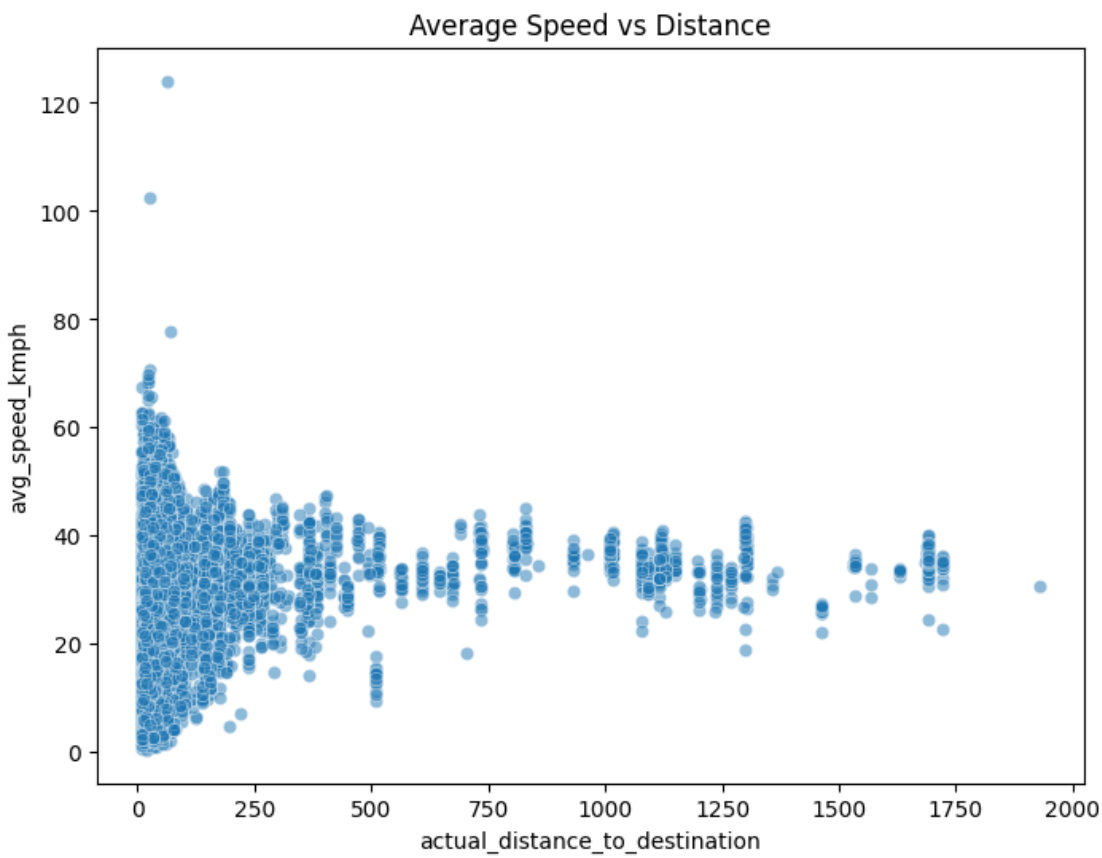
These strong correlations confirm that delivery distance and estimated route metrics are key predictors of delivery time. Therefore, these variables will be highly useful for building predictive models.

Average Speed vs Distance

```
[ ]: plt.figure(figsize=(8,6))

     sns.scatterplot(x='actual_distance_to_destination',␣
      ↪y='avg_speed_kmph',data=trip,alpha=0.5)

     plt.title("Average Speed vs Distance")
     plt.show()
```



- Observation:Average delivery speed generally ranges between 20–40 km/h, regardless of distance.
- Business Insight: The relatively consistent speed range suggests that operational constraints such as traffic conditions and logistics processes maintain similar travel speeds across different routes.

Overall EDA Insight The exploratory analysis reveals that delivery time is strongly influenced by travel distance and routing estimates. Most deliveries occur within short distances and moderate time ranges, while a small number of trips experience significant delays. Major logistics activity is concentrated around metropolitan cities and industrial hubs such as Mumbai, Bengaluru, and Gurgaon. These insights highlight key operational patterns that can help improve delivery forecasting and route optimization.

1.9.13 Outlier Detection & Treatment

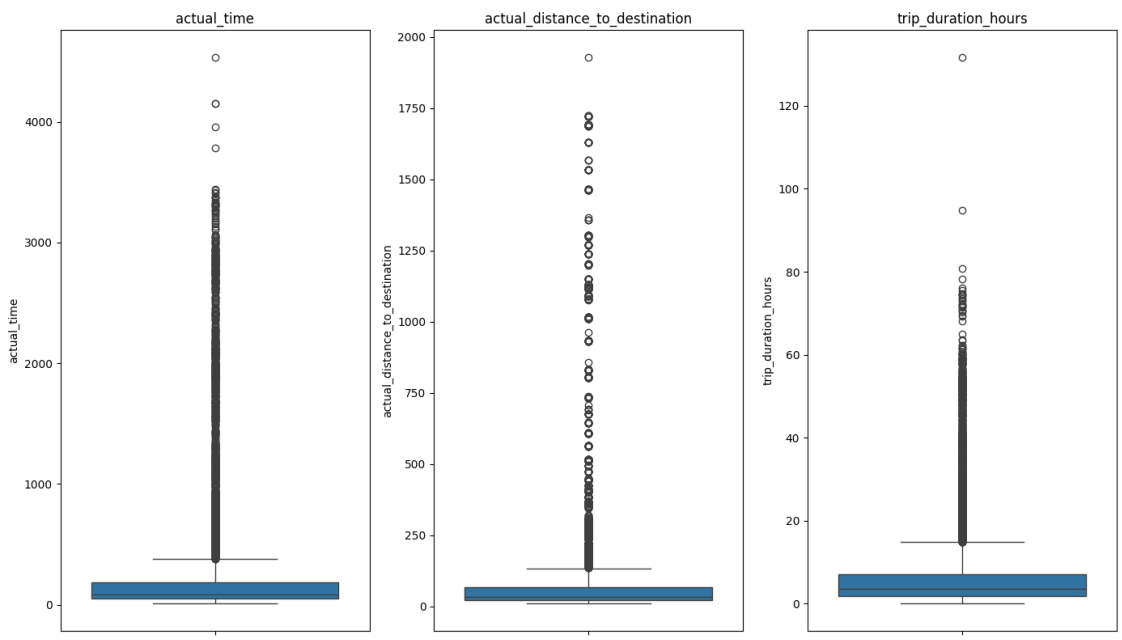
Outliers are extreme values that deviate significantly from the rest of the data. These values can distort statistical analysis and negatively impact machine learning models. Therefore, it is important to identify and handle outliers in numerical features.

[ ]:
```
# Visualizing Outliers using Boxplots
cols_for_outliers = [
    'actual_time',
    'actual_distance_to_destination',
    'trip_duration_hours'
]

plt.figure(figsize=(14,8))

for i,col in enumerate(cols_for_outliers):
    plt.subplot(1,3,i+1)
    sns.boxplot(y=trip[col])
    plt.title(col)

plt.tight_layout()
plt.show()
```



Observation

The boxplots indicate the presence of several extreme values in delivery time, delivery distance, and trip duration. These values lie beyond the upper whiskers of the boxplots, suggesting that some trips take significantly longer or travel much farther than the majority of deliveries.

Business Insight

The presence of these extreme values may be attributed to:

- Long-distance interstate shipments
- Operational delays in logistics hubs
- Traffic congestion and route inefficiencies
- Possible data recording inconsistencies

Such extreme values can distort statistical analysis and machine learning models, making it necessary to treat them appropriately.

Handling Outliers using IQR Method

[ ]:
```
cols_for_outliers = [
    'actual_time',
    'actual_distance_to_destination',
    'trip_duration_hours'
]

trip_clean = trip.copy()

for col in cols_for_outliers:

    Q1 = trip_clean[col].quantile(0.25)
    Q3 = trip_clean[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    trip_clean = trip_clean[(trip_clean[col] >= lower) & (trip_clean[col] <=␣
 ↪upper)]

print("Original:", trip.shape)
print("After cleaning:", trip_clean.shape)
```

```
Original: (13896, 27)
After cleaning: (10327, 27)
```

Since logistics operations may naturally contain long-distance deliveries, outlier treatment was applied cautiously only to critical operational variables to avoid removing valid long-haul shipments.

Observation

Using the IQR method, extreme values were filtered from the dataset. After removing outliers from key operational variables, the dataset size reduced from 13,896 rows to 10,327 rows.

Insight

The outlier removal process helps ensure that the dataset better represents typical delivery operations, while eliminating unusually large values that could bias analysis and predictive modeling.

1.9.14 One - Hot Encoding

[ ]:
```
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler , MinMaxScaler , OneHotEncoder

cat_cols = ['data','route_type']
```

[ ]:
```
ohe = OneHotEncoder(sparse_output=False)
encoded_cat_cols = ohe.fit_transform(trip_clean[cat_cols])
encoded_df = pd.DataFrame(encoded_cat_cols,columns=ohe.
 ↪get_feature_names_out(cat_cols),index=trip_clean.index)
encoded_df
```

[ ]:
```
       data_test  data_training  route_type_Carting  route_type_FTL
1            0.0            1.0                 1.0             0.0
3            0.0            1.0                 1.0             0.0
4            0.0            1.0                 0.0             1.0
5            0.0            1.0                 1.0             0.0
6            0.0            1.0                 1.0             0.0
…          …            …                   …              …
14782        1.0            0.0                 1.0             0.0
14783        1.0            0.0                 1.0             0.0
14784        1.0            0.0                 1.0             0.0
14785        1.0            0.0                 1.0             0.0
14786        1.0            0.0                 0.0             1.0

[10327 rows x 4 columns]
```

[ ]:
```
trip_encoded = pd.concat([trip_clean, encoded_df], axis=1)

print("Shape after encoding:", trip_encoded.shape)
trip_encoded.head()
```

```
Shape after encoding: (10327, 31)
```

[ ]:
```
                 trip_uuid      data  trip_creation_time  \
1  trip-153671042288605164  training 2018-09-12 00:00:23
3  trip-153671046011330457  training 2018-09-12 00:01:00
4  trip-153671052974046625  training 2018-09-12 00:02:10
5  trip-153671055416136166  training 2018-09-12 00:02:34
6  trip-153671066201138152  training 2018-09-12 00:04:22

                                 route_schedule_uuid route_type source_center  \
1  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…    Carting  IND561203AAB
```

```
3  thanos::sroute:f0176492-a679-4597-8332-bbd1c7f…    Carting  IND400072AAB
4  thanos::sroute:d9f07b12-65e0-4f3b-bec8-df06134…        FTL  IND583101AAA
5  thanos::sroute:9bf03170-d0a2-4a3f-aa4d-9aaab3d…    Carting  IND600056AAA
6  thanos::sroute:a97698cc-846e-41a7-916b-88b1741…    Carting  IND600044AAD

  destination_center       od_start_time         od_end_time  \
1       IND561203AAB 2018-09-12 02:03:10 2018-09-12 02:03:10
3       IND401104AAA 2018-09-12 00:01:00 2018-09-12 01:41:30
4       IND583119AAA 2018-09-12 00:02:10 2018-09-12 03:54:43
5       IND600056AAA 2018-09-12 02:12:11 2018-09-12 02:12:11
6       IND600048AAA 2018-09-12 00:04:22 2018-09-12 01:42:22

   actual_distance_to_destination  …  source_city  destination_city  \
1                       48.542890  …   Doddablpur        Doddablpur
3                       17.175274  …  Mumbai Hub             Mumbai
4                       26.600536  …      Bellary            Sandur
5                        9.271519  …      Chennai           Chennai
6                        9.100510  …      Chennai           Chennai

   source_state  destination_state  avg_speed_kmph  time_error  data_test  \
1     Karnataka          Karnataka       30.339306          54        0.0
3   Maharashtra        Maharashtra       17.466381          44        0.0
4     Karnataka          Karnataka       25.333844          36        0.0
5    Tamil Nadu         Tamil Nadu       26.490056          10        0.0
6    Tamil Nadu         Tamil Nadu       22.751275          11        0.0

   data_training  route_type_Carting  route_type_FTL
1            1.0                 1.0             0.0
3            1.0                 1.0             0.0
4            1.0                 0.0             1.0
5            1.0                 1.0             0.0
6            1.0                 1.0             0.0

[5 rows x 31 columns]
```

[ ]:
```
trip_temp = trip_encoded.drop(columns=['data', 'route_type', 'data_test',␣
↪'route_type_Carting'])
trip_temp.head()
```

[ ]:
```
                 trip_uuid  trip_creation_time  \
1  trip-153671042288605164 2018-09-12 00:00:23
3  trip-153671046011330457 2018-09-12 00:01:00
4  trip-153671052974046625 2018-09-12 00:02:10
5  trip-153671055416136166 2018-09-12 00:02:34
6  trip-153671066201138152 2018-09-12 00:04:22

                                  route_schedule_uuid source_center  \
```

```
1  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…  IND561203AAB
3  thanos::sroute:f0176492-a679-4597-8332-bbd1c7f…  IND400072AAB
4  thanos::sroute:d9f07b12-65e0-4f3b-bec8-df06134…  IND583101AAA
5  thanos::sroute:9bf03170-d0a2-4a3f-aa4d-9aaab3d…  IND600056AAA
6  thanos::sroute:a97698cc-846e-41a7-916b-88b1741…  IND600044AAD

  destination_center       od_start_time         od_end_time  \
1       IND561203AAB 2018-09-12 02:03:10 2018-09-12 02:03:10
3       IND401104AAA 2018-09-12 00:01:00 2018-09-12 01:41:30
4       IND583119AAA 2018-09-12 00:02:10 2018-09-12 03:54:43
5       IND600056AAA 2018-09-12 02:12:11 2018-09-12 02:12:11
6       IND600048AAA 2018-09-12 00:04:22 2018-09-12 01:42:22

   actual_distance_to_destination  actual_time  osrm_time  …  trip_weekday  \
1                       48.542890           96         42  …             2
3                       17.175274           59         15  …             2
4                       26.600536           63         27  …             2
5                        9.271519           21         11  …             2
6                        9.100510           24         13  …             2

   trip_hour  source_city  destination_city  source_state  destination_state  \
1          0   Doddablpur        Doddablpur     Karnataka          Karnataka
3          0  Mumbai Hub             Mumbai   Maharashtra        Maharashtra
4          0      Bellary            Sandur     Karnataka          Karnataka
5          0      Chennai           Chennai    Tamil Nadu         Tamil Nadu
6          0      Chennai           Chennai    Tamil Nadu         Tamil Nadu

   avg_speed_kmph  time_error  data_training route_type_FTL
1       30.339306          54            1.0            0.0
3       17.466381          44            1.0            0.0
4       25.333844          36            1.0            1.0
5       26.490056          10            1.0            0.0
6       22.751275          11            1.0            0.0

[5 rows x 27 columns]
```

Categorical features were transformed using One-Hot Encoding to convert them into numerical form suitable for machine learning algorithms. Redundant dummy variables were removed to prevent multicollinearity.

1.9.15 Feature Scaling

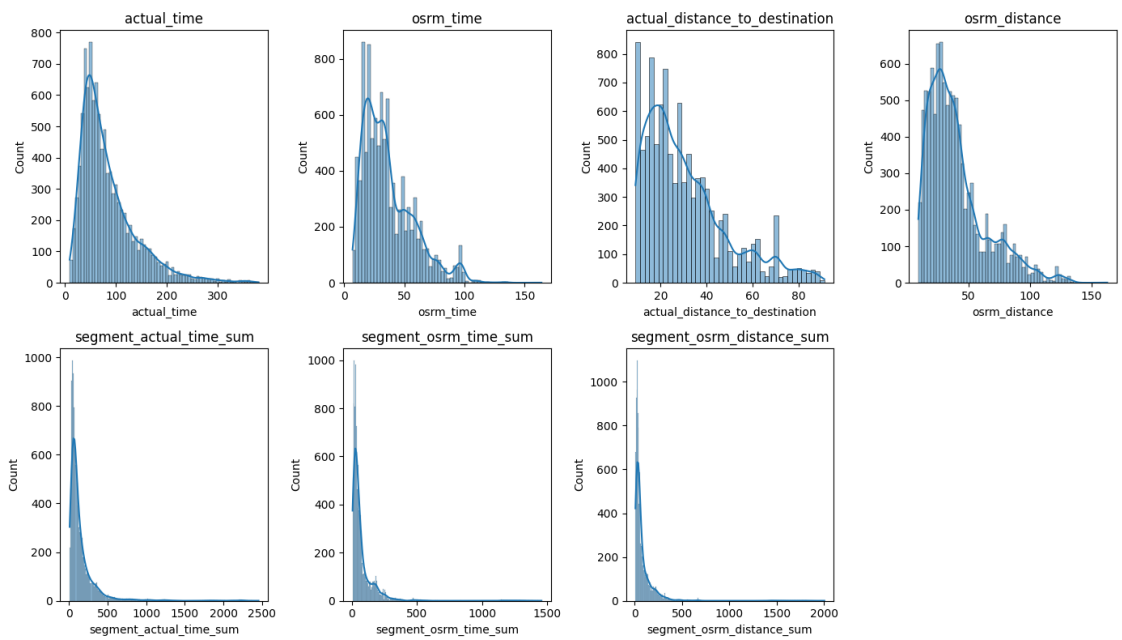
[ ]:
```
num_cols = [
'actual_time',
'osrm_time',
'actual_distance_to_destination',
'osrm_distance',
```

```
'segment_actual_time_sum',
'segment_osrm_time_sum',
'segment_osrm_distance_sum'
]

plt.figure(figsize=(14,8))

for i,col in enumerate(num_cols):
    plt.subplot(2,4,i+1)
    sns.histplot(trip_clean[col], kde=True)
    plt.title(col)

plt.tight_layout()
plt.show()
```



The histograms show that most numerical variables are right-skewed, with a large concentration of observations at lower values and a long tail extending toward higher values. This skewness indicates that the dataset does not follow a Gaussian distribution, which influences the choice of feature scaling technique.

Minmax scaler - Most appropriate since the data is not gaussian MinMaxScaler transforms features to a range between 0 and 1, which helps machine learning models perform better when features have different scales.

[ ]:
```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
```

```
trip_temp[num_cols] = scaler.fit_transform(trip_temp[num_cols])

trip_temp.head()
```

[ ]:

```
                 trip_uuid  trip_creation_time  \
1  trip-153671042288605164 2018-09-12 00:00:23
3  trip-153671046011330457 2018-09-12 00:01:00
4  trip-153671052974046625 2018-09-12 00:02:10
5  trip-153671055416136166 2018-09-12 00:02:34
6  trip-153671066201138152 2018-09-12 00:04:22

                                route_schedule_uuid source_center  \
1  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…  IND561203AAB
3  thanos::sroute:f0176492-a679-4597-8332-bbd1c7f…  IND400072AAB
4  thanos::sroute:d9f07b12-65e0-4f3b-bec8-df06134…  IND583101AAA
5  thanos::sroute:9bf03170-d0a2-4a3f-aa4d-9aaab3d…  IND600056AAA
6  thanos::sroute:a97698cc-846e-41a7-916b-88b1741…  IND600044AAD

  destination_center       od_start_time         od_end_time  \
1       IND561203AAB 2018-09-12 02:03:10 2018-09-12 02:03:10
3       IND401104AAA 2018-09-12 00:01:00 2018-09-12 01:41:30
4       IND583119AAA 2018-09-12 00:02:10 2018-09-12 03:54:43
5       IND600056AAA 2018-09-12 02:12:11 2018-09-12 02:12:11
6       IND600048AAA 2018-09-12 00:04:22 2018-09-12 01:42:22

   actual_distance_to_destination  actual_time  osrm_time  …  trip_weekday  \
1                        0.480588     0.233871   0.222930  …             2
3                        0.099335     0.134409   0.050955  …             2
4                        0.213893     0.145161   0.127389  …             2
5                        0.003270     0.032258   0.025478  …             2
6                        0.001192     0.040323   0.038217  …             2

   trip_hour  source_city  destination_city  source_state  destination_state  \
1          0   Doddablpur        Doddablpur     Karnataka          Karnataka
3          0  Mumbai Hub             Mumbai   Maharashtra        Maharashtra
4          0      Bellary            Sandur     Karnataka          Karnataka
5          0      Chennai           Chennai    Tamil Nadu         Tamil Nadu
6          0      Chennai           Chennai    Tamil Nadu         Tamil Nadu

   avg_speed_kmph  time_error  data_training route_type_FTL
1       30.339306          54            1.0            0.0
3       17.466381          44            1.0            0.0
4       25.333844          36            1.0            1.0
5       26.490056          10            1.0            0.0
6       22.751275          11            1.0            0.0
```

```
[5 rows x 27 columns]
```

- Observation: After applying MinMax scaling, all numerical variables are transformed into a range between 0 and 1, ensuring consistent feature scales.
- Insight: Since the dataset is not normally distributed, MinMaxScaler is more appropriate than StandardScaler, as it scales the data without assuming normality. This prevents features with larger numerical ranges from dominating the model.

1.10 Hypothesis Testing

1. Are aggregated actual_time and aggregated osrm_time similar? Setting Significance level = 0.05

STEP-1 : Set up Null Hypothesis & Alternate Hypothesis

- Null Hypothesis (H0) - There is no significant difference in the mean values between column1 and column2
- Alternate Hypothesis(HA) - There is a significant difference in the mean values between column1 and column2

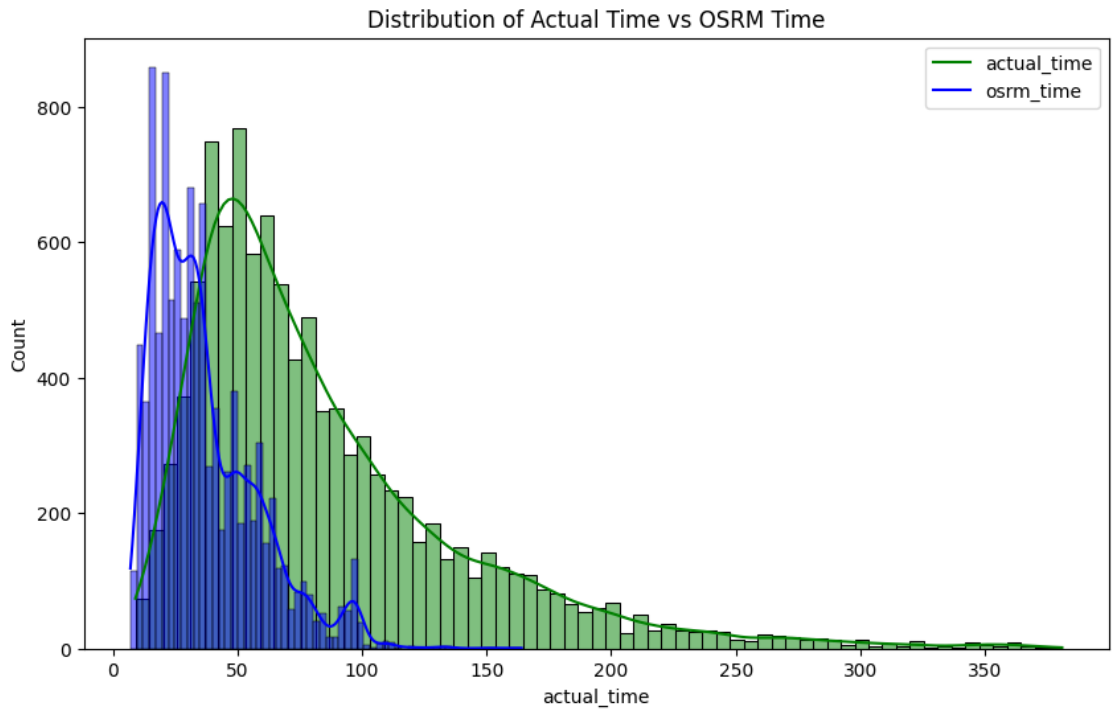
Step-2 : Visual Analysis

[ ]:
```
plt.figure(figsize=(10,6))

sns.histplot(trip_clean['actual_time'], color='green', kde=True)
sns.histplot(trip_clean['osrm_time'], color='blue', kde=True)

plt.legend(['actual_time','osrm_time'])
plt.title("Distribution of Actual Time vs OSRM Time")

plt.show()
```



The distribution of actual delivery time is shifted toward higher values compared to OSRM estimated time, indicating that the routing engine tends to underestimate delivery durations.

Since your dataset contains over 10,000 observations, the Central Limit Theorem applies.

This means that even if the difference is not perfectly normal, the paired t-test remains reliable for large samples.

Step-3 : Perform test

[ ]:
```
from scipy.stats import ttest_rel

t_stat , p_value = ttest_rel(trip_clean['actual_time'],trip_clean['osrm_time'])

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: 107.74248559261038
P-value: 0.0
```

[ ]:
```
def accept_reject(p_value, alpha=0.05):
    if p_value > alpha:
        print(f'Fail to reject null hypothesis (p-value = {p_value:.4f} >␣
 ↪{alpha})')
        print('Result is NOT statistically significant')
    else:
```

```
        print(f'Reject null hypothesis (p-value = {p_value:.4f} <= {alpha})')
        print('Result IS statistically significant')
```

[ ]:
```
accept_reject(p_value)
```

```
Reject null hypothesis (p-value = 0.0000 <= 0.05)
Result IS statistically significant
```

This indicates that actual delivery times are significantly different from OSRM estimated times.

In most cases, actual delivery time is higher, suggesting that the routing engine does not fully account for real-world factors such as:

- Traffic congestion
- Loading/unloading delays
- Warehouse processing time
- Route disruptions

2. Are aggregated actual_time and aggregated segment_actual_time_sum similar?

Step 1 — Define Hypotheses

- Null Hypothesis (H): There is no significant difference between the mean of actual_time and segment_actual_time_sum.
- Alternative Hypothesis (H): There is a significant difference between the mean of actual_time and segment_actual_time_sum.

Step 2 - Visual Analysis

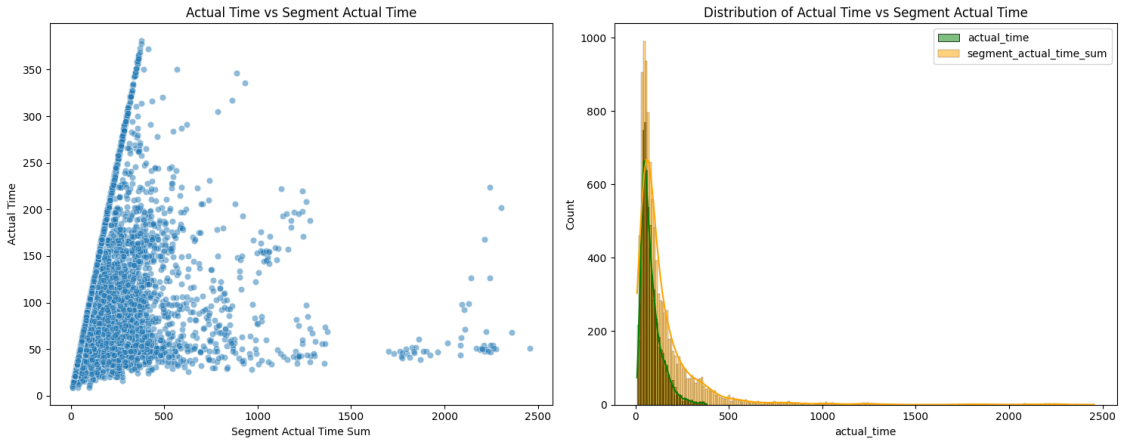
[ ]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

# First subplot - Scatter plot
sns.
 ↪scatterplot(x='segment_actual_time_sum',y='actual_time',data=trip_clean,alpha=0.
 ↪5,ax=ax1 )
ax1.set_title("Actual Time vs Segment Actual Time")
ax1.set_xlabel("Segment Actual Time Sum")
ax1.set_ylabel("Actual Time")

# Second subplot - Histogram with KDE
sns.histplot(trip_clean['actual_time'], color='green', kde=True, ax=ax2,␣
 ↪label='actual_time')
sns.histplot(trip_clean['segment_actual_time_sum'], color='orange', kde=True,␣
 ↪ax=ax2, label='segment_actual_time_sum')
ax2.set_title("Distribution of Actual Time vs Segment Actual Time")
ax2.legend()

plt.tight_layout()
```

```
plt.show()
```



The distribution of segment_actual_time_sum appears shifted toward higher values compared to actual_time, indicating that the sum of segment travel times is often larger than the total recorded trip time.

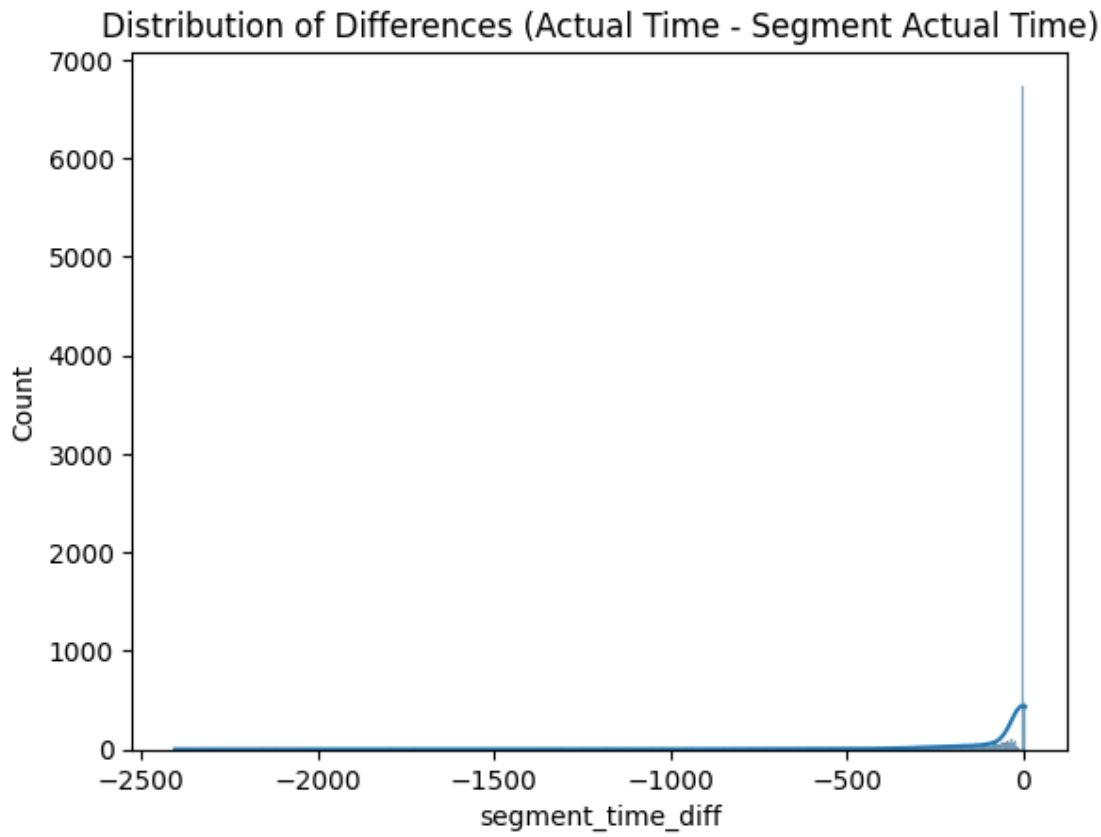
A positive relationship exists between segment-level actual time and total trip delivery time.

Step 3 - Check Assumption (Normality of Differences)

[ ]:
```
trip_clean['segment_time_diff'] = (trip_clean['actual_time'] -␣
↪trip_clean['segment_actual_time_sum'])

sns.histplot(trip_clean['segment_time_diff'], kde=True)

plt.title("Distribution of Differences (Actual Time - Segment Actual Time)")
plt.show()
```



Although the distribution may not be perfectly normal, the dataset contains a large number of observations, so the Central Limit Theorem allows the paired t-test to remain valid.

Step 4 — Perform Paired T-Test

[ ]:
```
t_stat , p_value =␣
↪ttest_rel(trip_clean['actual_time'],trip_clean['segment_actual_time_sum'])

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: -35.459787050319996
P-value: 5.2902043148433084e-260
```

[ ]:
```
accept_reject(p_value)
```

```
Reject null hypothesis (p-value = 0.0000 <= 0.05)
Result IS statistically significant
```

The p-value obtained from the paired t-test is less than 0.05, indicating a statistically significant difference between actual_time and segment_actual_time_sum.

Insight

This suggests that the sum of segment travel times differs from the recorded total trip time. The difference may arise because segment-level times include additional operational activities such as scanning, loading, and hub processing, which may not directly contribute to the overall trip duration.

3. Are aggregated osrm_distance and segment_osrm_distance_sum similar?

Step 1 — Define Hypotheses

- Null Hypothesis (H): There is no significant difference between the mean of osrm_distance and segment_osrm_distance_sum.
- Alternative Hypothesis (H): There is a significant difference between the mean of osrm_distance and segment_osrm_distance_sum.

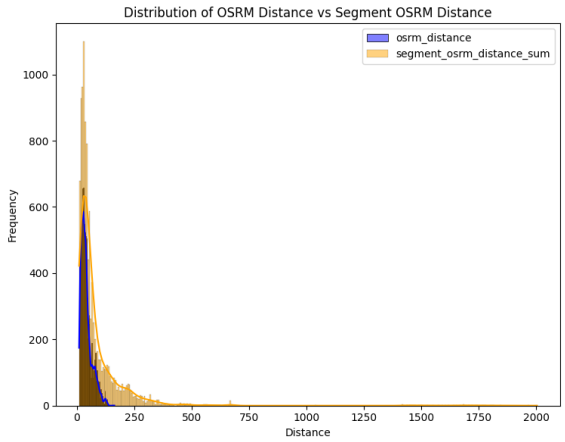
Step 2 - Visual Analysis

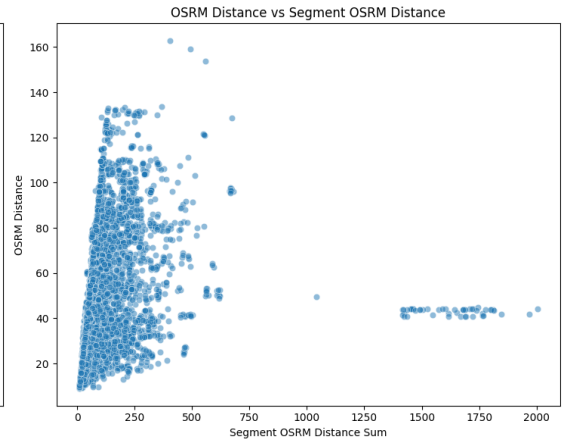
[ ]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

# First subplot - Histogram with KDE
sns.histplot(trip_clean['osrm_distance'], color='blue', kde=True, ax=ax1,␣
↪label='osrm_distance')
sns.histplot(trip_clean['segment_osrm_distance_sum'], color='orange', kde=True,␣
↪ax=ax1, label='segment_osrm_distance_sum')
ax1.set_title("Distribution of OSRM Distance vs Segment OSRM Distance")
ax1.legend()
ax1.set_xlabel("Distance")
ax1.set_ylabel("Frequency")

# Second subplot - Scatter plot
sns.scatterplot(
    x='segment_osrm_distance_sum',
    y='osrm_distance',
    data=trip_clean,
    alpha=0.5,
    ax=ax2
)
ax2.set_title("OSRM Distance vs Segment OSRM Distance")
ax2.set_xlabel("Segment OSRM Distance Sum")
ax2.set_ylabel("OSRM Distance")

# Adjust layout to prevent overlapping
plt.tight_layout()
plt.show()
```





Both variables show similar distribution patterns, suggesting that the total OSRM route distance is closely related to the sum of segment-level OSRM distances.

The scatter plot shows a strong positive linear relationship between segment_osrm_distance_sum and osrm_distance.

Step 4 — Perform Paired T-Test

[ ]:
```
t_stat , p_value =␣
↪ttest_rel(trip_clean['osrm_distance'],trip_clean['segment_osrm_distance_sum'])

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: -34.410152975261305
P-value: 9.477019935173543e-246
```

[ ]:
```
accept_reject(p_value)
```

```
Reject null hypothesis (p-value = 0.0000 <= 0.05)
Result IS statistically significant
```

The p-value obtained from the paired t-test is less than 0.05, indicating a statistically significant difference between osrm_distance and segment_osrm_distance_sum.

The slight difference between aggregated OSRM distance and the sum of segment-level OSRM distances may arise due to route adjustments, intermediate stops, or rounding differences in distance calculations. However, the strong relationship confirms that segment-level routing estimates are largely consistent with overall route distance predictions.

4. Are aggregated osrm_time and segment_osrm_time_sum similar?

Step 1 — Define Hypotheses

- Null Hypothesis (H): There is no significant difference between the mean of osrm_time and segment_osrm_time_sum.

- Alternative Hypothesis (H): There is a significant difference between the mean of osrm_time and segment_osrm_time_sum.

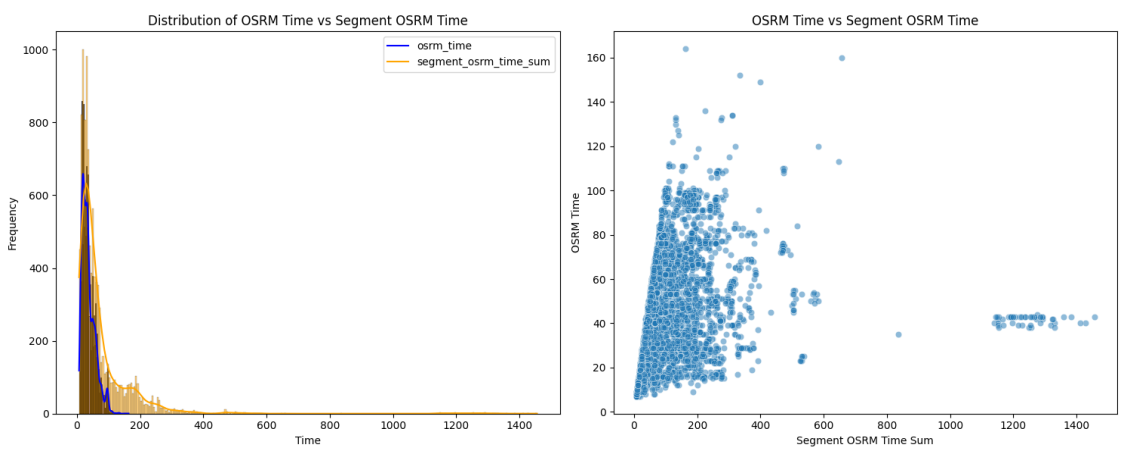
Step 2 - Visual Analysis

[ ]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

# First subplot - Histogram with KDE
sns.histplot(trip_clean['osrm_time'], color='blue', kde=True, ax=ax1,␣
↪label='osrm_time')
sns.histplot(trip_clean['segment_osrm_time_sum'], color='orange', kde=True,␣
↪ax=ax1, label='segment_osrm_time_sum')
ax1.set_title("Distribution of OSRM Time vs Segment OSRM Time")
ax1.legend(['osrm_time', 'segment_osrm_time_sum'])
ax1.set_xlabel("Time")
ax1.set_ylabel("Frequency")

# Second subplot - Scatter plot
sns.scatterplot(x='segment_osrm_time_sum',y='osrm_time',data=trip_clean,alpha=0.
↪5,ax=ax2)
ax2.set_title("OSRM Time vs Segment OSRM Time")
ax2.set_xlabel("Segment OSRM Time Sum")
ax2.set_ylabel("OSRM Time")

# Adjust layout to prevent overlapping
plt.tight_layout()
plt.show()
```



Both variables display similar distribution shapes, indicating that the total OSRM travel time is closely related to the sum of segment-level OSRM travel times.

The scatter plot shows a strong positive linear relationship between segment_osrm_time_sum and osrm_time.

Step 3 — Perform Paired T-Test

[ ]:
```
t_stat , p_value =␣
↪ttest_rel(trip_clean['osrm_time'],trip_clean['segment_osrm_time_sum'])

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: -37.79925760053928
P-value: 5.828985359199367e-293
```

[ ]:
```
accept_reject(p_value)
```

```
Reject null hypothesis (p-value = 0.0000 <= 0.05)
Result IS statistically significant
```

The p-value obtained from the paired t-test is less than 0.05, indicating a statistically significant difference between osrm_time and segment_osrm_time_sum.

Insight:

The difference between aggregated OSRM travel time and the sum of segment-level OSRM times may arise due to routing adjustments, intermediate hub delays, or estimation approximations in the routing engine. Nevertheless, the strong relationship confirms that segment-level routing predictions are generally consistent with the overall route travel time estimate.

1.11 Business Insight

- The dataset covers deliveries between 12 September 2018 and 8 October 2018, representing approximately 26 days of logistics operations.
- Most numerical variables such as actual_time, osrm_time, and distance features are right-skewed, indicating that while most deliveries occur within normal operational ranges, a smaller number of trips take significantly longer due to long-distance routes or operational delays.
- Delivery time and distance show strong positive correlations, indicating that distance is one of the primary drivers of delivery duration.
- Hypothesis testing results indicate that actual delivery time differs significantly from OSRM estimated time. This suggests that the routing engine systematically underestimates delivery time, likely because it does not fully account for:
  - traffic congestion
  - loading and unloading delays
  - warehouse processing time
  - route diversions
- The top source and destination states are: Maharashtra, Karnataka, and Haryana. This indicates that a large portion of logistics activity is concentrated in major economic hubs and industrial corridors.

- The cities with the highest shipment activity include: Bengaluru, Mumbai, and Gurgaon. These cities likely act as major logistics hubs in the Delhivery network.
- Hypothesis testing between aggregated values and segment-level values shows that:
  - segment times and distances are strongly correlated with trip-level metrics
  - however, small statistical differences exist due to operational activities between segments

This confirms that segment aggregation logic is largely correct.

- Carting shipments account for a large share of deliveries, indicating that many shipments involve multi-stop local delivery operations rather than full truckload shipments.

1.12 Business Recommendation

- Since OSRM estimated time significantly differs from actual delivery time, Delhivery should enhance routing models by incorporating: traffic patterns, warehouse handling time, and regional congestion trends. This would improve delivery time prediction accuracy.
- States such as Maharashtra, Karnataka, Haryana, Tamil Nadu, Telangana, and Gujarat show high shipment activity. Optimizing logistics infrastructure and route planning in these corridors could significantly improve operational efficiency.
- Differences between segment-level travel time and total trip time suggest possible delays at intermediate hubs. Improving hub processing time could reduce overall delivery duration.
- Differences between segment-level travel time and total trip time suggest possible delays at intermediate hubs. Improving hub processing time could reduce overall delivery duration.
- The routing engine should continuously update routes using real-time data, including traffic patterns and historical delivery performance.
- Since delivery time strongly depends on distance, incorporating distance-based predictive models could improve delivery time forecasting accuracy.

Conclusion

The analysis highlights that delivery time is strongly influenced by travel distance and operational factors such as routing efficiency and hub processing time. While OSRM provides useful route estimates, the observed differences between estimated and actual delivery times suggest opportunities for improving prediction accuracy. By optimizing high-traffic corridors, enhancing routing algorithms, and improving hub-level operations, Delhivery can significantly improve delivery efficiency and customer satisfaction.

[ ]:

```
```